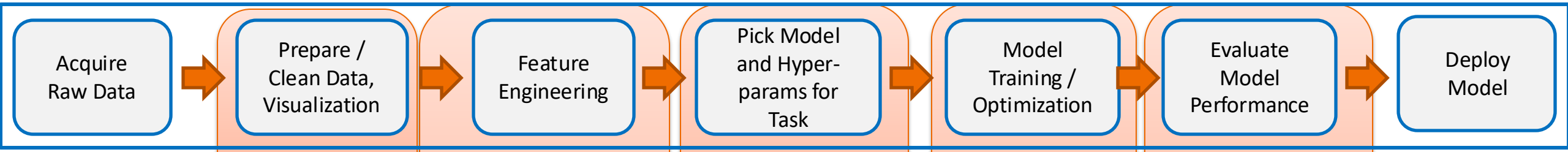
Focus of this lecture



- Visualization

- Representing Word2Vec, Image: Visual BoW, deep network features
- Representing Audio : Spectrogram, Mel features
- Dimensionality Reduction (PCA)

- Multi-Layer Perceptron(MLP)

- Gradient Descent
- Support Vector machines
- Clustering

- Performance Measures (Accuracy, Precision, Recall, F-1)

Today's Plan

Session 1:

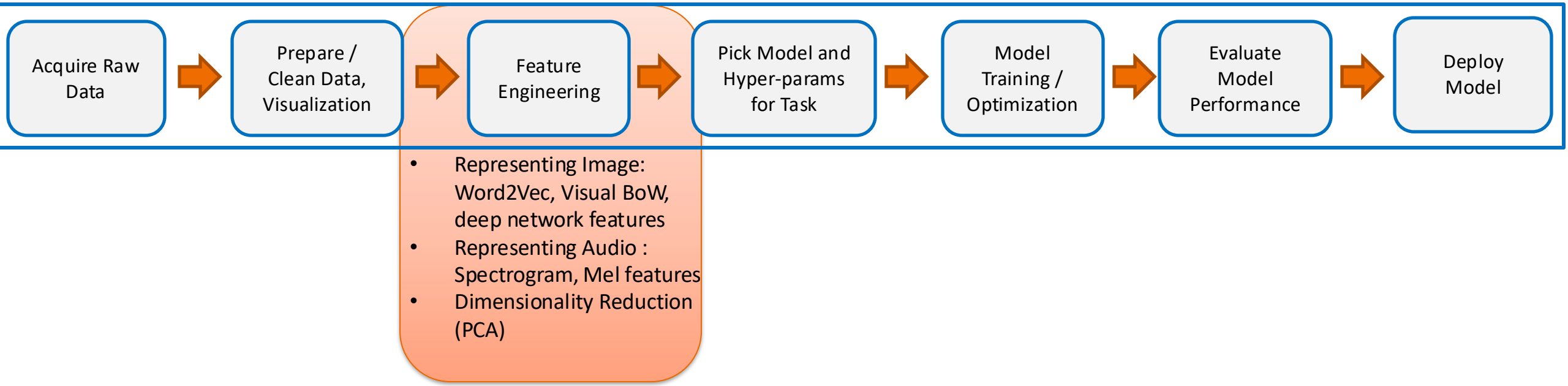
1. Representation
   - Word2Vec
   - Image: Visual BoW, deep network features
   - Audio : Spectrogram, Mel features
2. Dimensionality Reduction (PCA)
3. Visualization
4. Performance Metrics

Session 2:

5. Gradient Descent
6. MLP
7. SVM with Kernels
8. Clustering

Session 3:

- Paper Reading



Feature Engineering

Word2vec - Intuition

- Marco saw a furry little wampimuk hiding in the tree

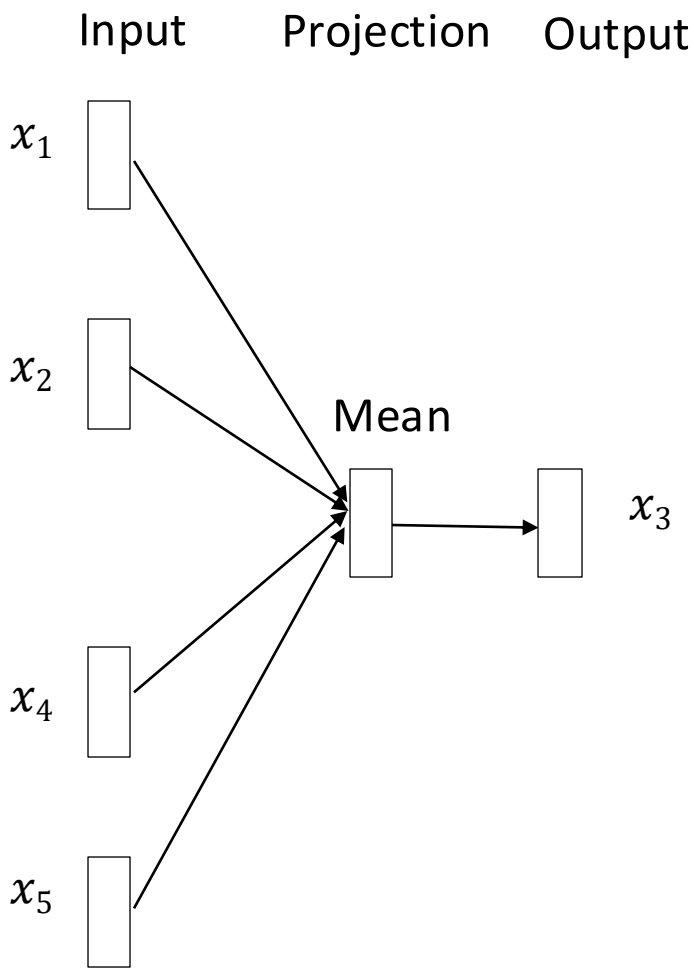
- **What is Marco?**
- **What is wampimuk?**

A person

Animal

Variants

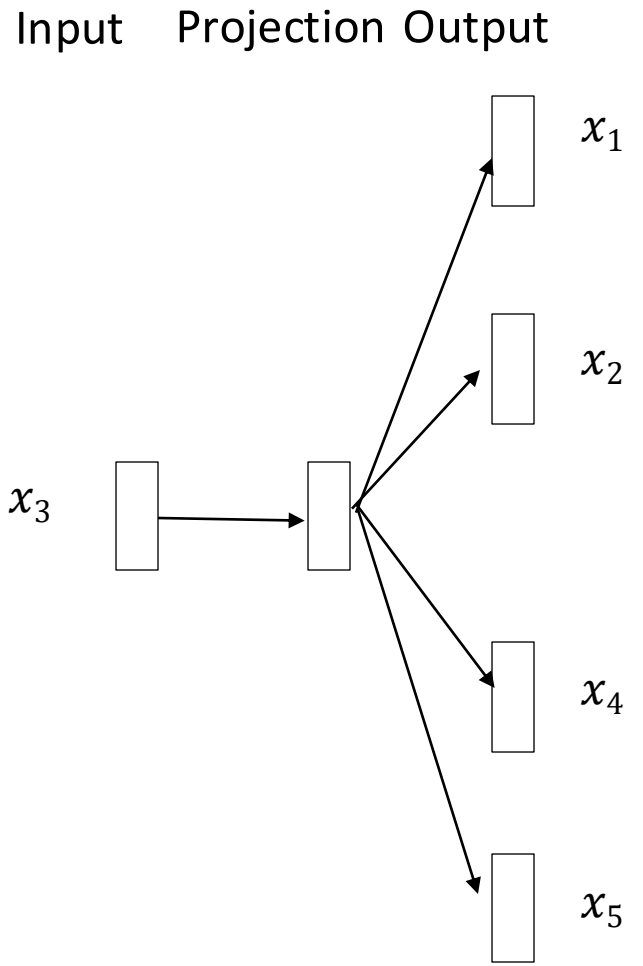
- Continuous Bag of Words (CBOW):
  - use a window of words to predict the missing word



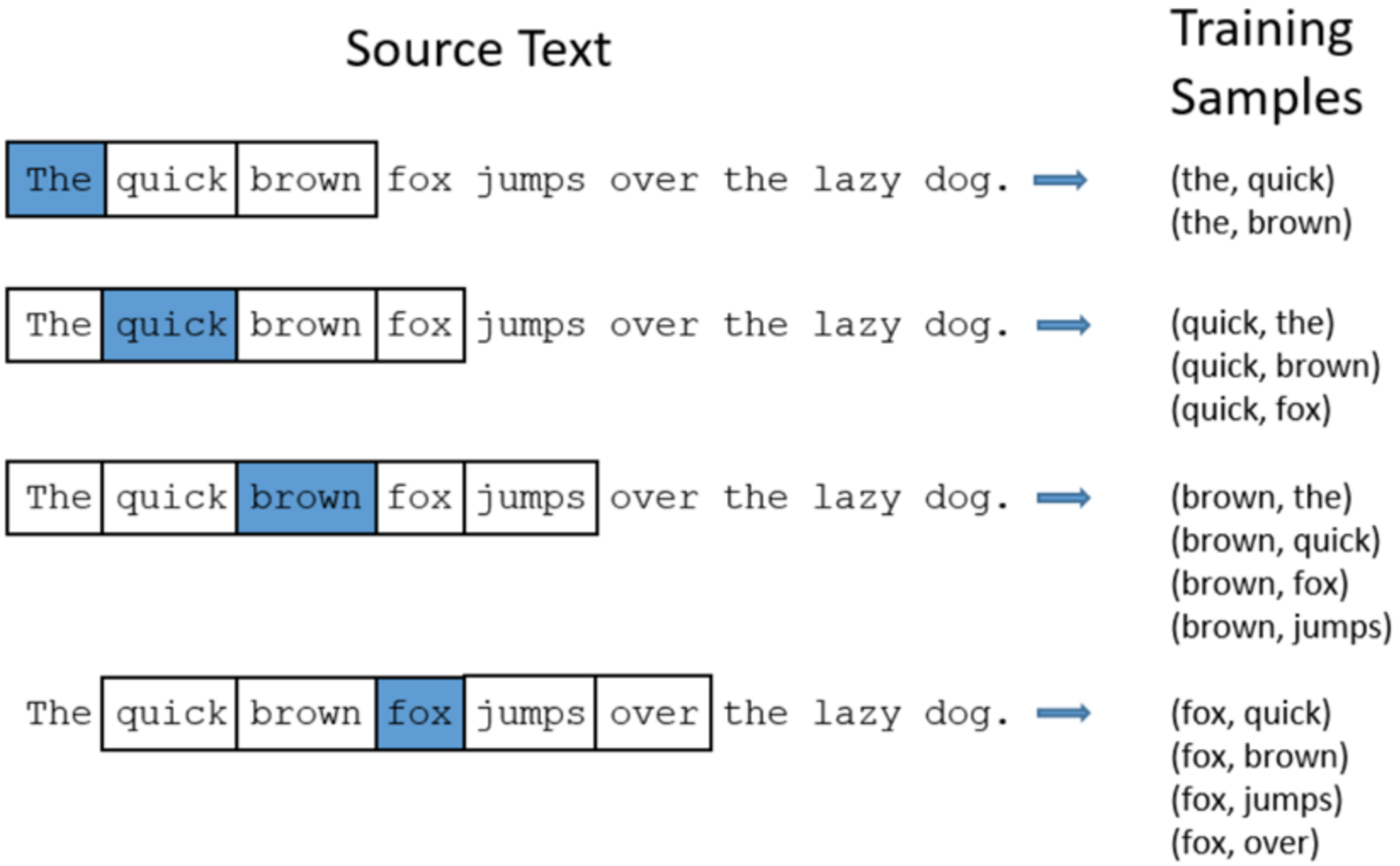
CBOW

Variants

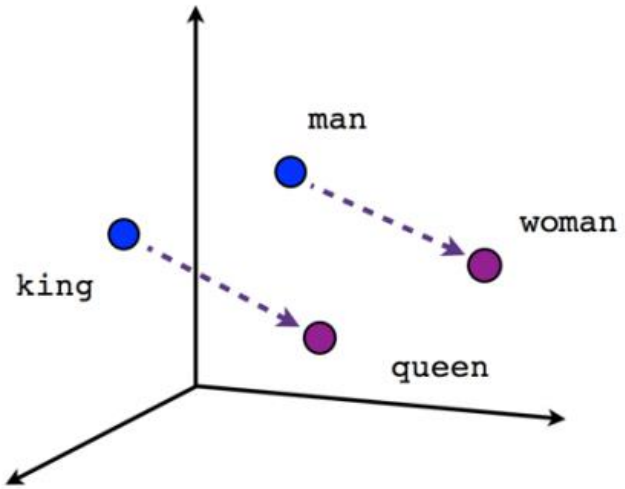
- Skip-gram (SG):
  - use a word to predict the surrounding ones in the specified window.



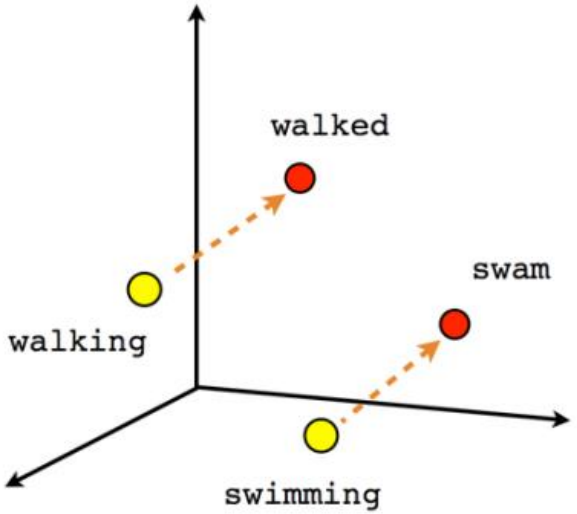
Word2vec – skip gram examples



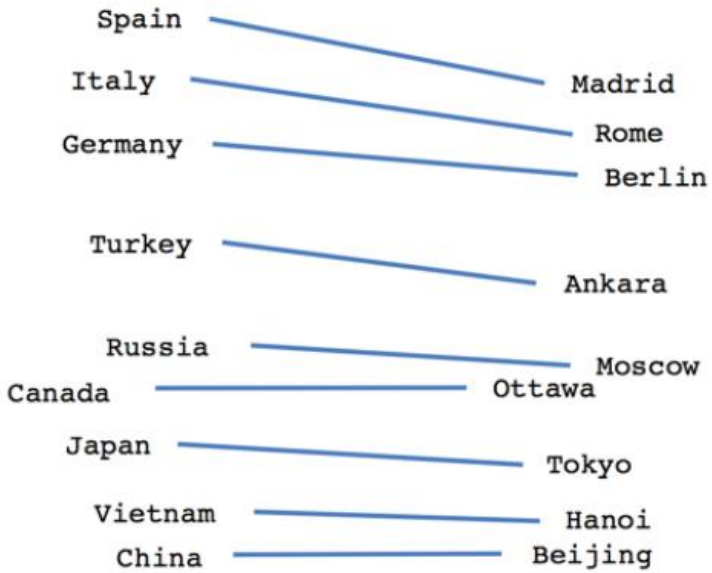
Examples



Male-Female



Verb tense



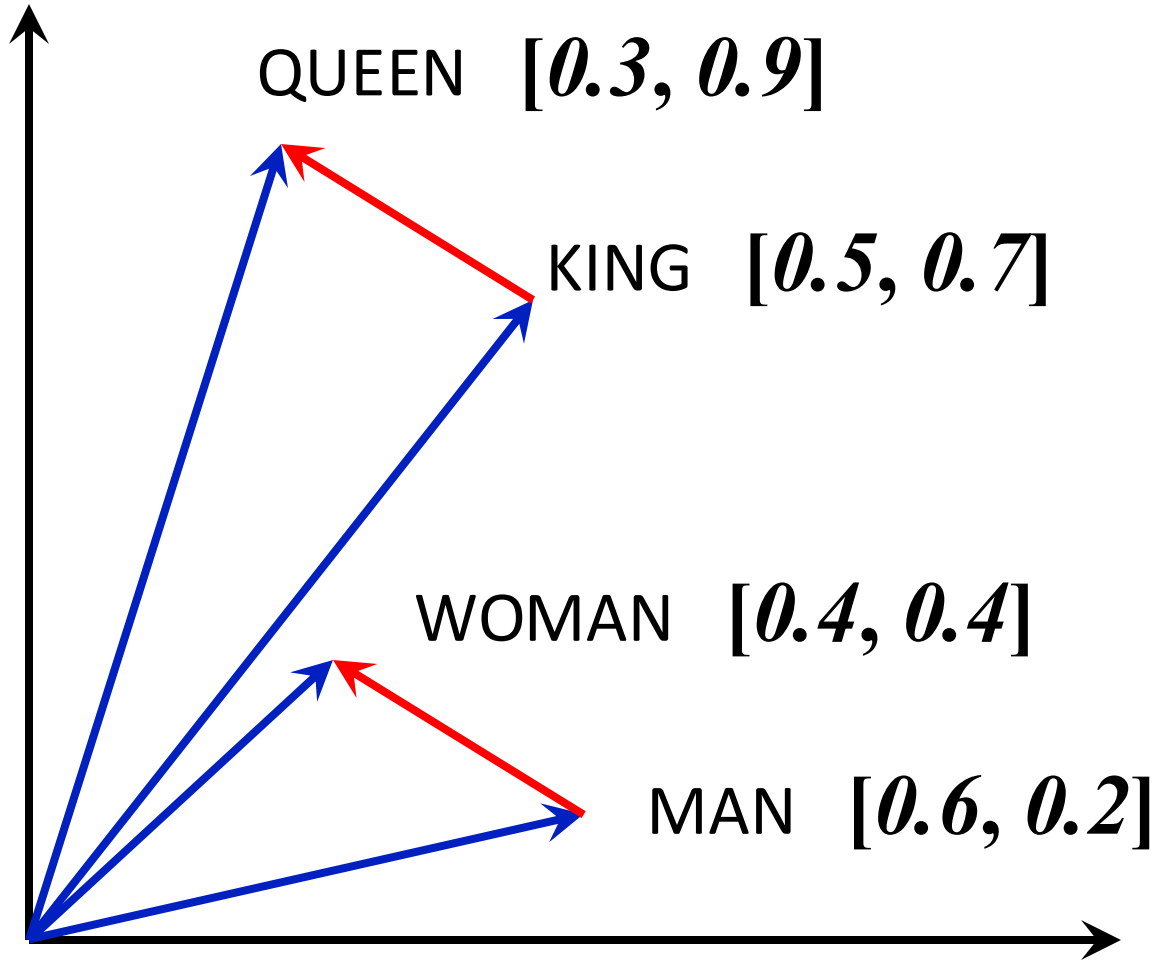
Country-Capital

https://www.tensorflow.org/images/linear-relationships.png

Visualising “Word-Arithmetic”

$$\boldsymbol{Vec(Queen)} = \boldsymbol{Vec(King)} - \boldsymbol{Vec(Man)} + \boldsymbol{Vec(Woman)}$$

- These fancy arithmetic were imagined (and shown) earlier also; but became popular with Word2Vec.



More Examples

FRANCE	JESUS	XBOX	REDDISH	SCRATCHED	MEGABITS
AUSTRIA	GOD	AMIGA	GREENISH	NAILED	OCTETS
BELGIUM	SATI	PLAYSTATION	BLUISH	SMASHED	MB/S
GERMANY	CHRIST	MSX	PINKISH	PUNCHED	BIT/S
ITALY	SATAN	IPOD	PURPLISH	POPPED	BAUD
GREECE	KALI	SEGA	BROWNISH	CRIMPED	CARATS
SWEDEN	INDRA	PSNUMBER	GREYISH	SCRAPED	KBIT/S
NORWAY	VISHNU	HD	GRAYISH	SCREWED	MEGAHERTZ
EUROPE	ANANDA	DREAMCAST	WHITISH	SECTIONED	MEGAPIXELS
HUNGARY	PARVATI	GEFORCE	SILVERY	SLASHED	GBIT/S
SWITZERLAND	GRACE	CAPCOM	YELLOWISH	RIPPED	AMPERES

What words have embeddings closest to a given word? From Collobert et al. (2011) (https://arxiv.org/pdf/1103.0398v1.pdf)

AI and Problem of Perception

Possible Features: Handcrafting

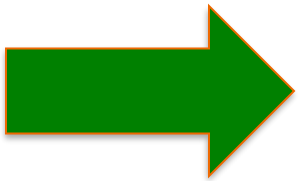
MIN RED
MAX RED
MEAN RED
MIN GREEN
MAX GREEN
MEAN GREEN
MIN BLUE
MAX BLUE
MEAN BLUE

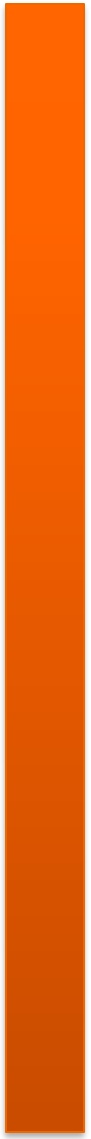
9 X 1
FEATURE VECTOR
PER IMAGE

Concerns:

- **Too naïve to capture the visual content?**
- **Too small to represent information?**

Possible Features: Raw Data Itself





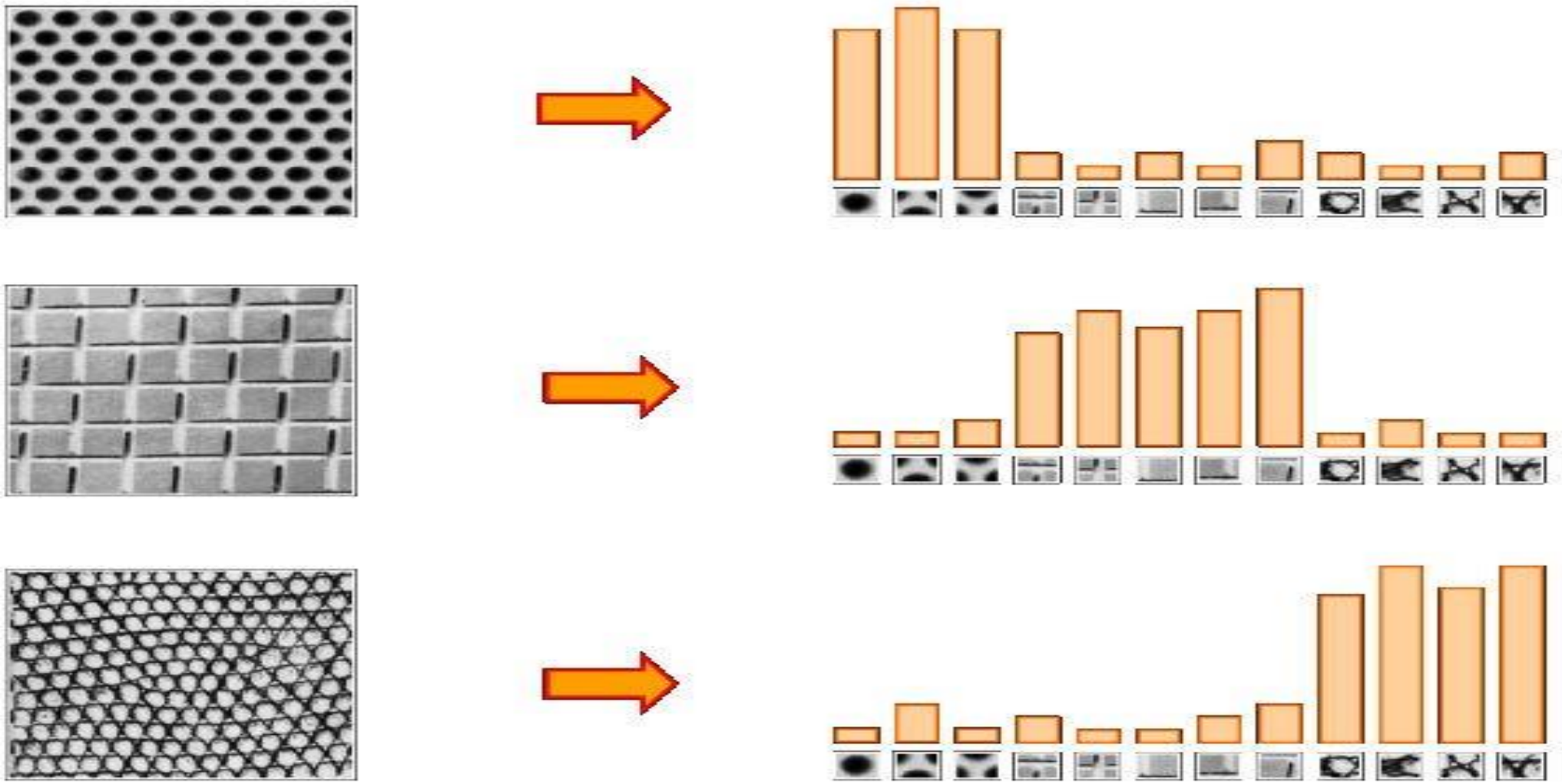
FEATURE VECTOR
32 X 32 X 3 = 3072
DIMENSION
PER IMAGE (d = 3072)

3072 X 1
vector

CONCERNS:

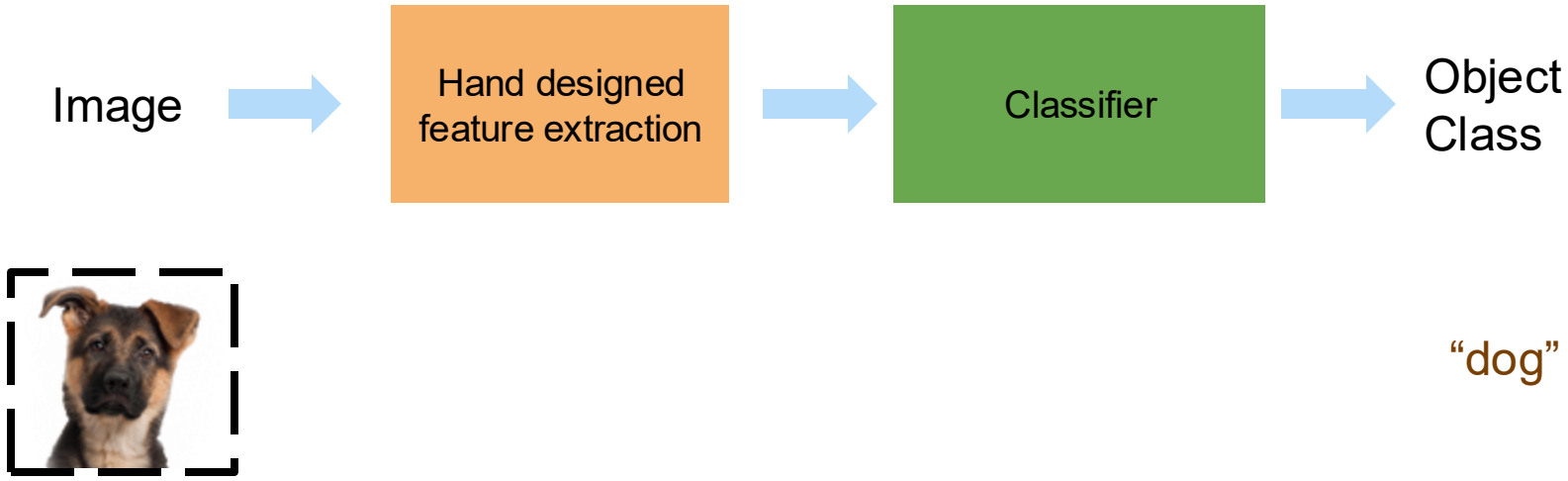
- Too big ?
- May be redundancy ?
- Too rigid?

Visual BoW: Basic Idea

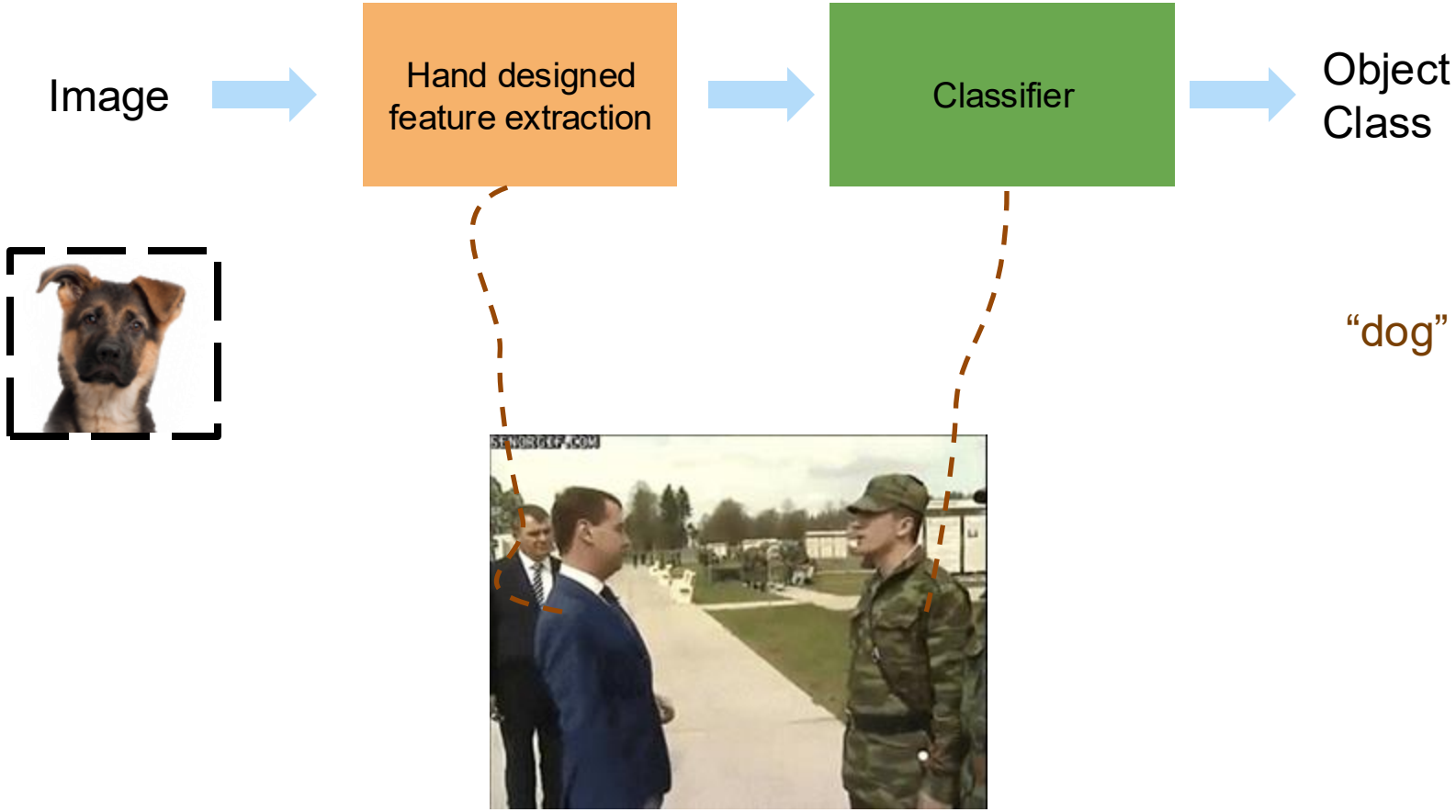


Source: wordpress.com/2013/04/24/research-bag-of-features-for-visual-recognition

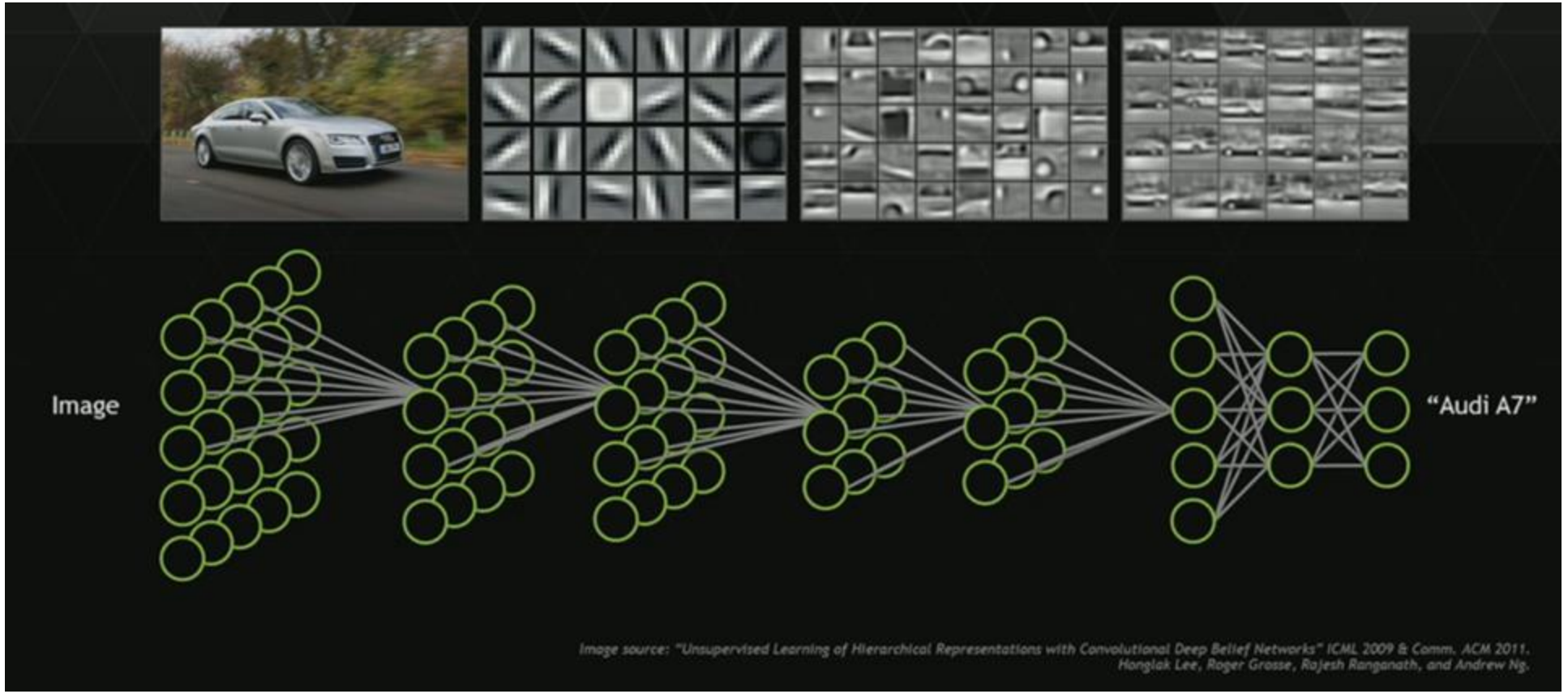
Object-recognition: conventional approach



Object-recognition: conventional approach



Object Recognition: Deep Neural Networks



Data-driven, End-to-End learning, Task-specific feature hierarchy

Summary

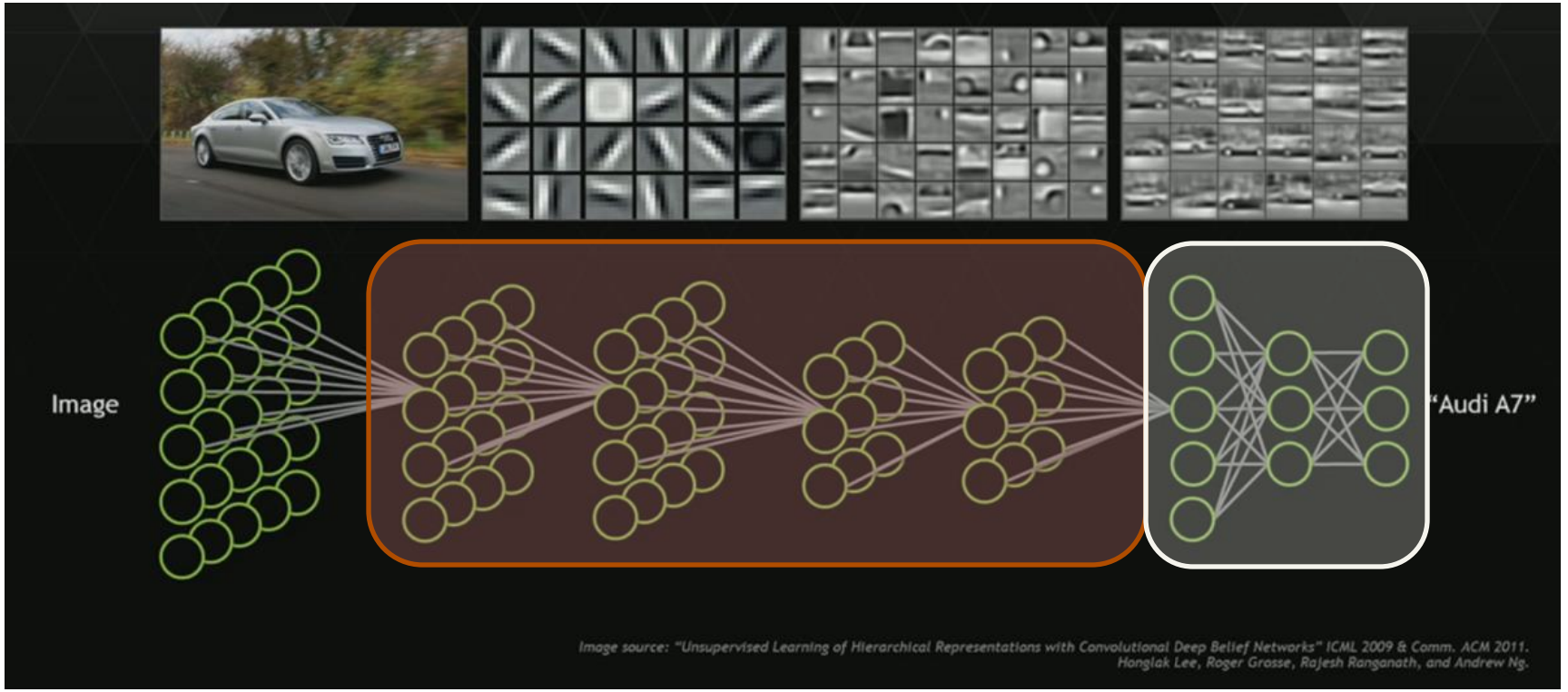
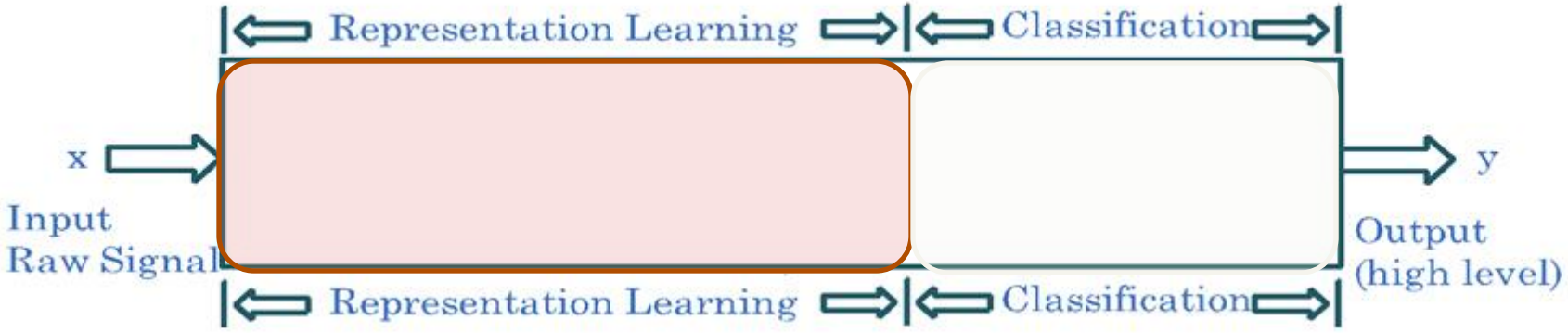
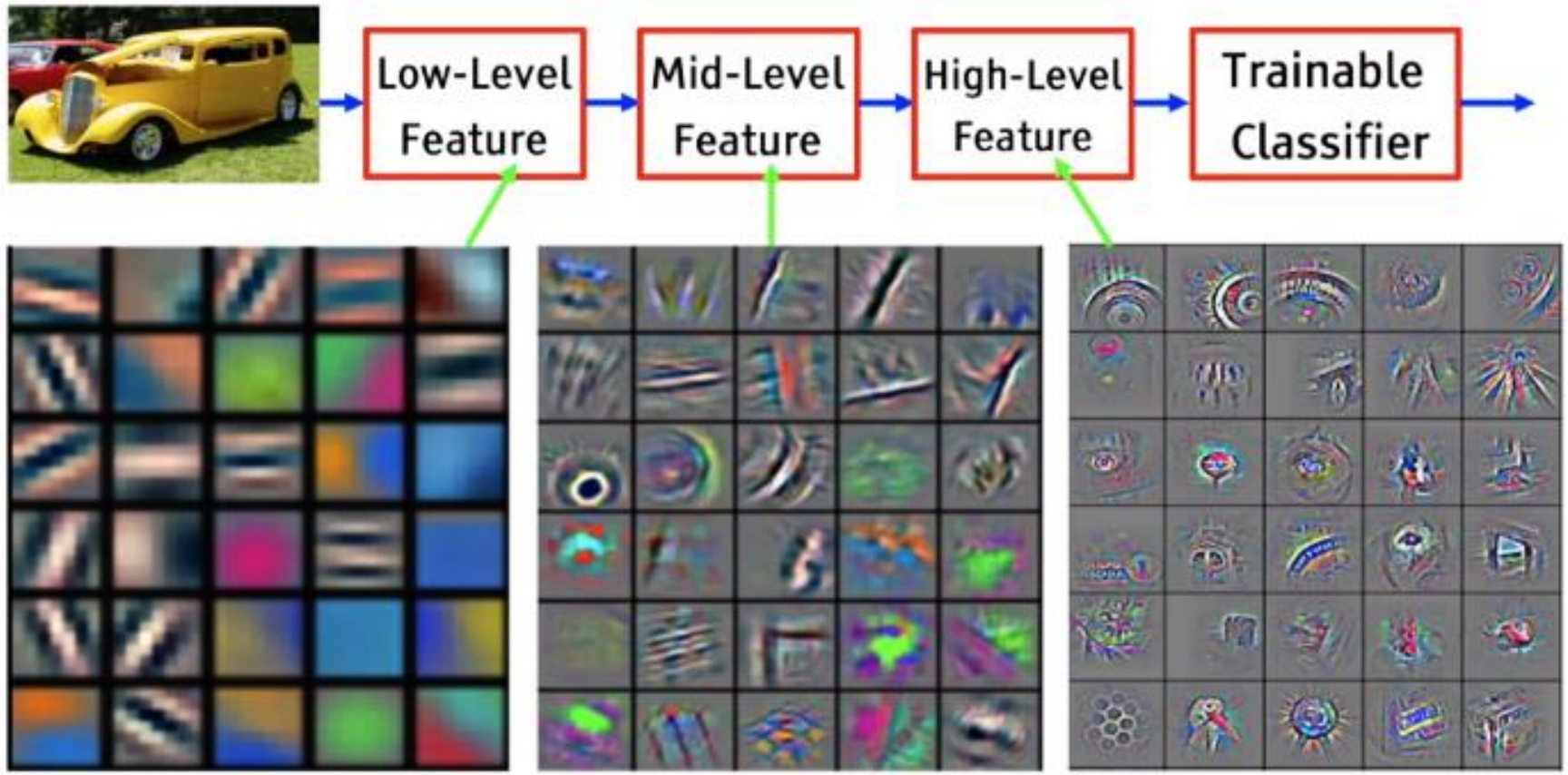


Image
"Audi A7"
Image source: "Unsupervised Learning of Hierarchical Representations with Convolutional Deep Belief Networks" ICML 2009 & Comm. ACM 2011.
Honglak Lee, Roger Grosse, Rajesh Ranganath, and Andrew Ng.

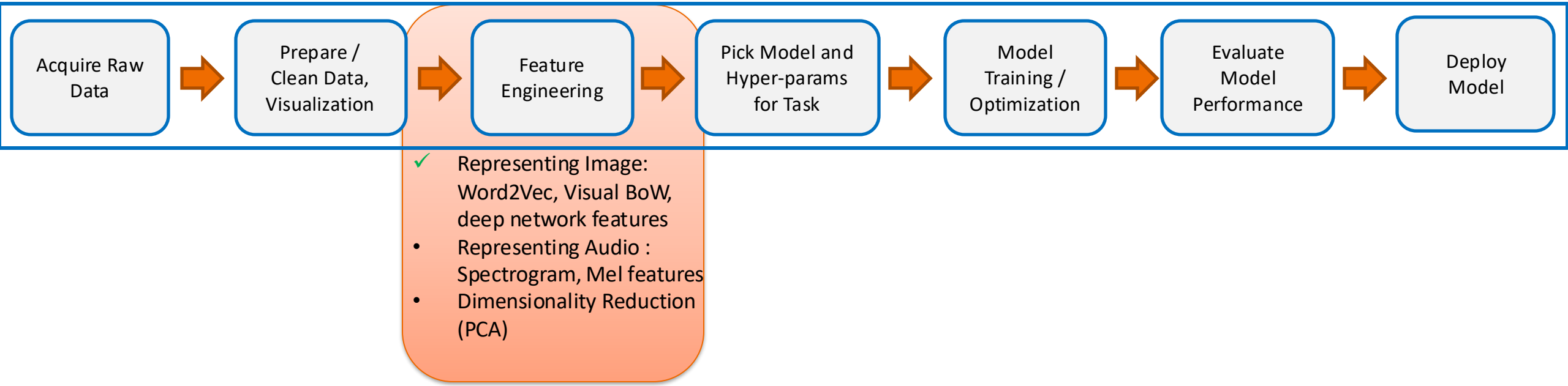

Representation Learning
Classification
x
Input
Raw Signal
y
Output
(high level)
Representation Learning
Classification

Deep Learnt Features (2013-XXX)

- It's deep if it has more than one stage of non-linear feature transformation.



Source: Yann LeCun

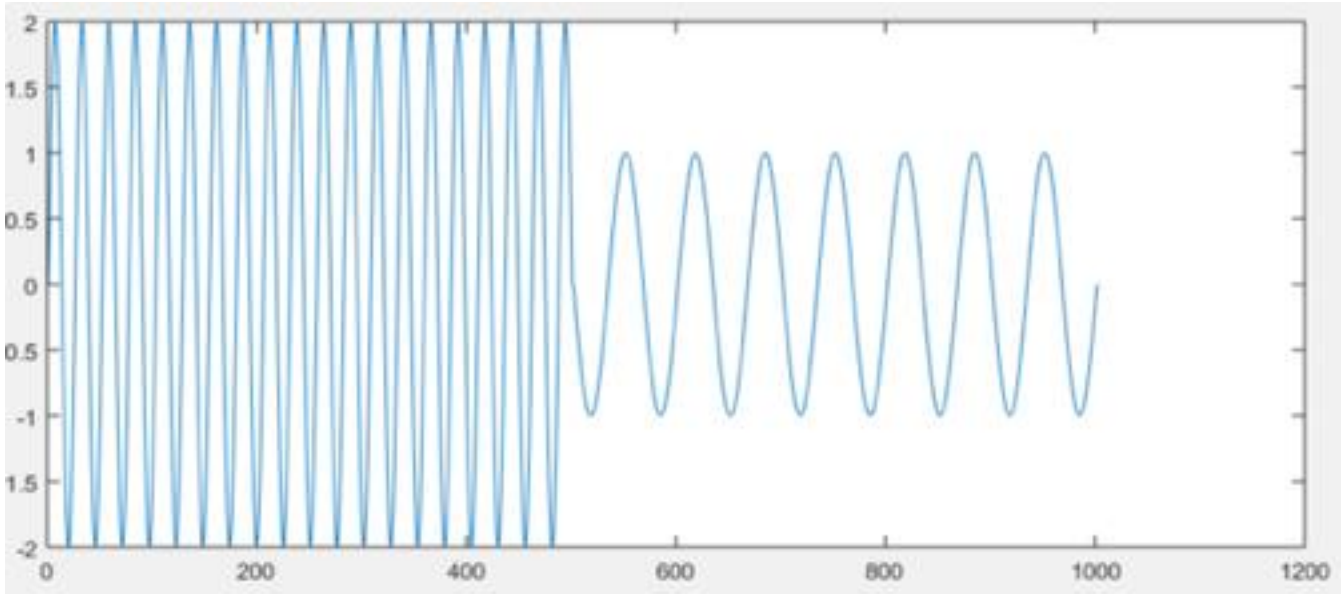


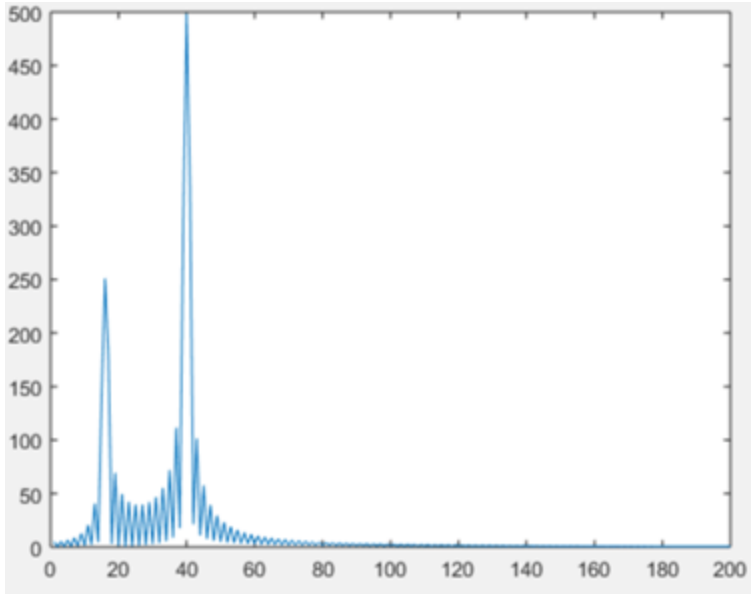
Speech

(Brief Explanation)

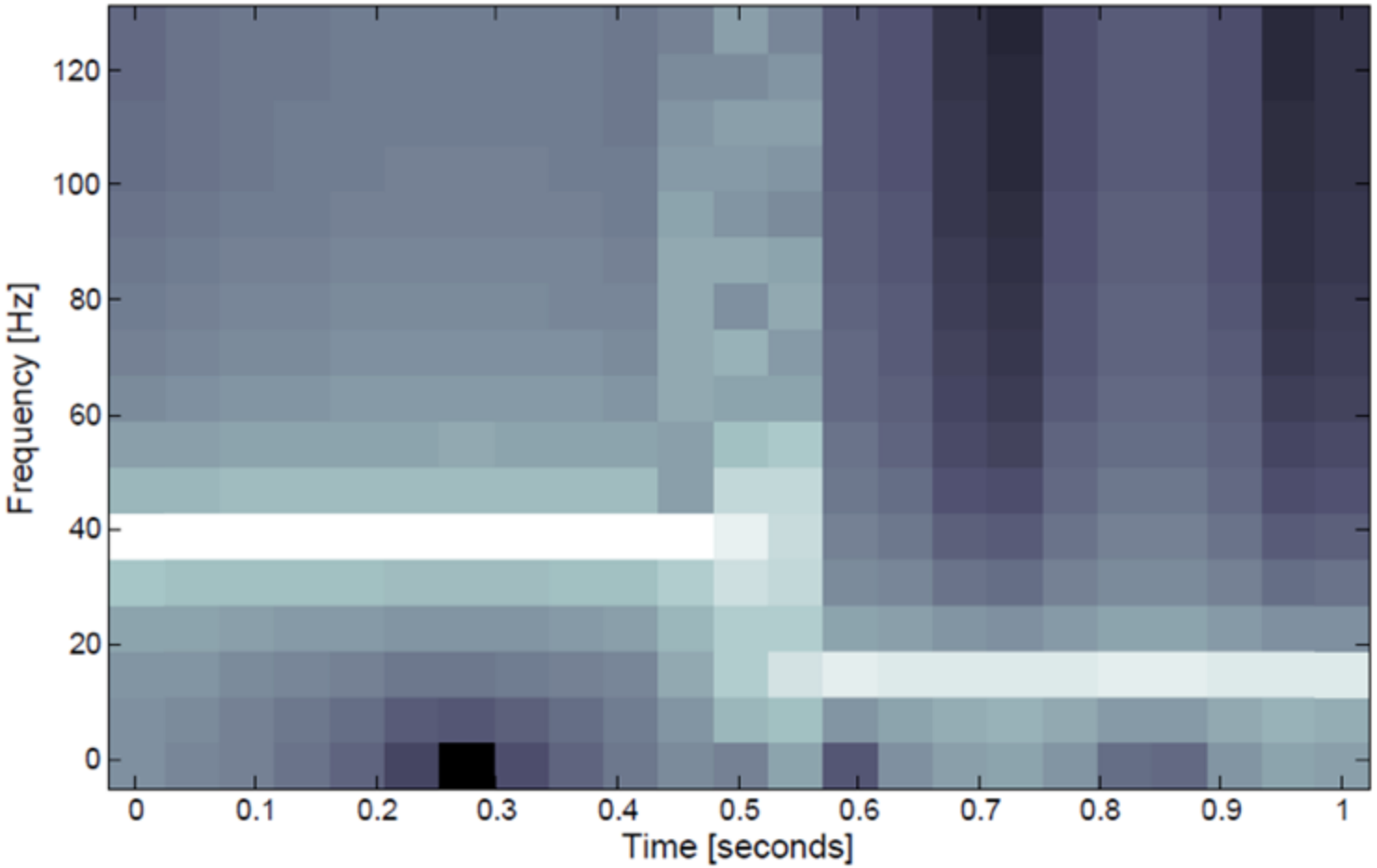
Example Sound Signal

$$g(t) = \begin{cases} 2 * \sin(2\pi \cdot 39t), 0 \leq t \leq 1/2 \\ \sin(2\pi \cdot 15t), 1/2 < t \leq 1 \end{cases}$$





Spectrogram



Spectrogram of a piecewise monochromatic signal.

Lighter color greater DFT magnitude

Representations

A: MFCC (Signal processing based; Classical)

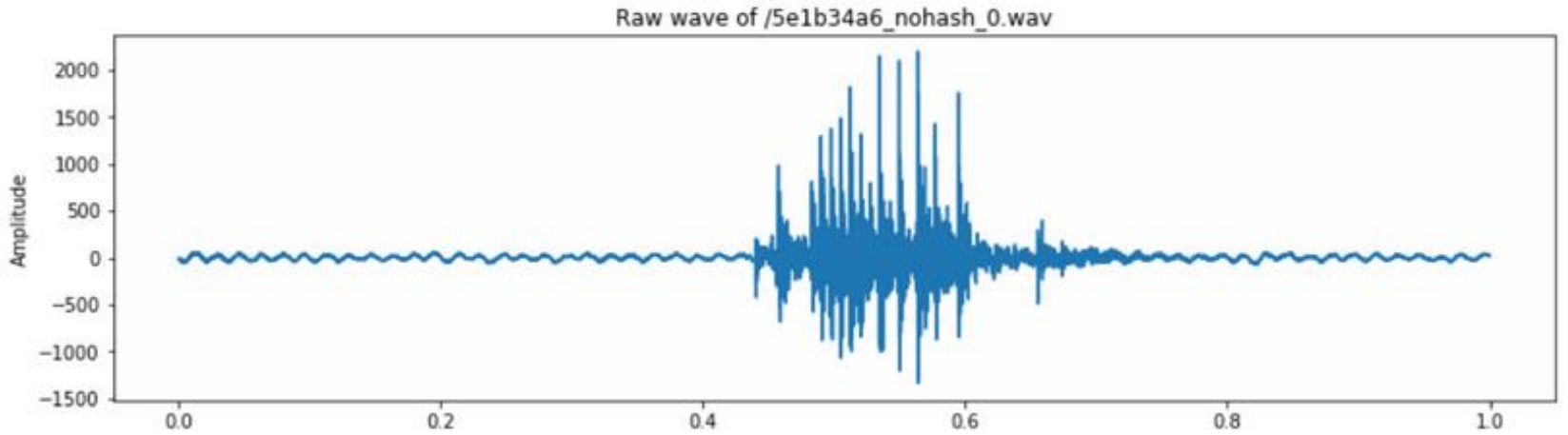
- Mel Frequency Cepstral Coefficients

B: CNN Based (Modern)

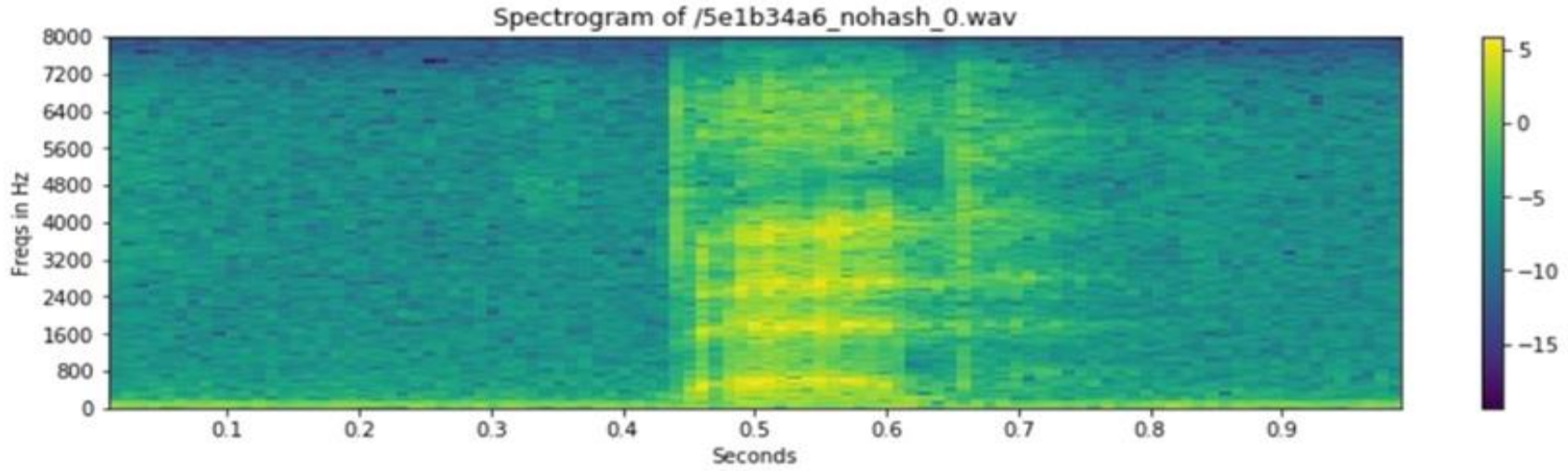
- VGG Features on the Mel Spectrogram

http://practicalcryptography.com/miscellaneous/machine-learning/guide-mel-frequency-cepstral-coefficients-mfccs/

Classical Feature (MFCC)

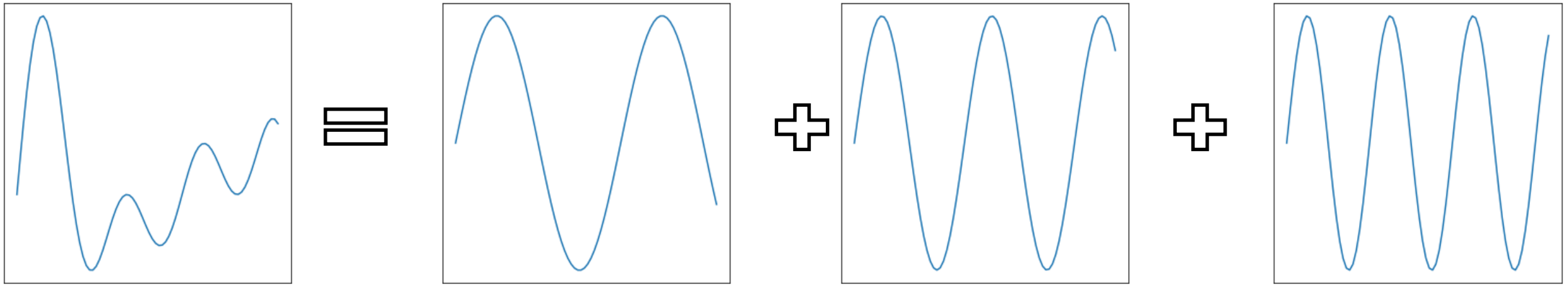


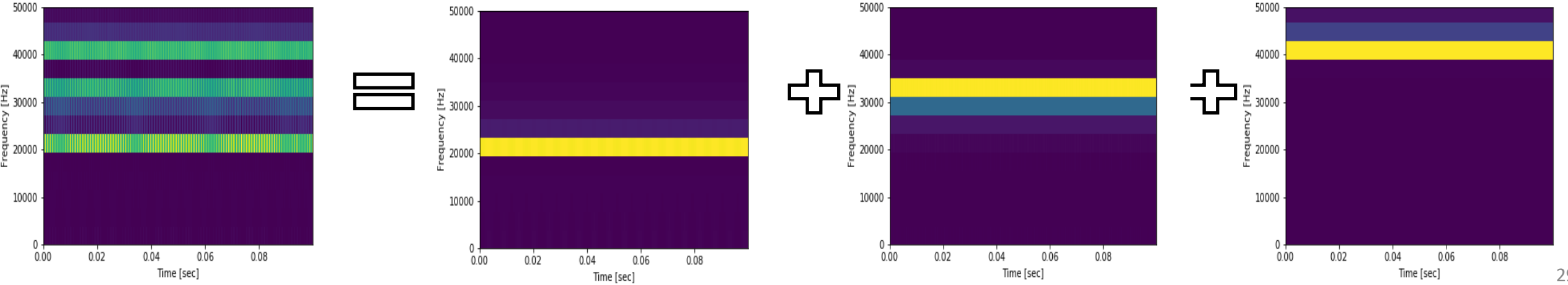
Amplitude
Vs
Time



Frequency
Vs
Time

Any wave is a combination of many sine wave

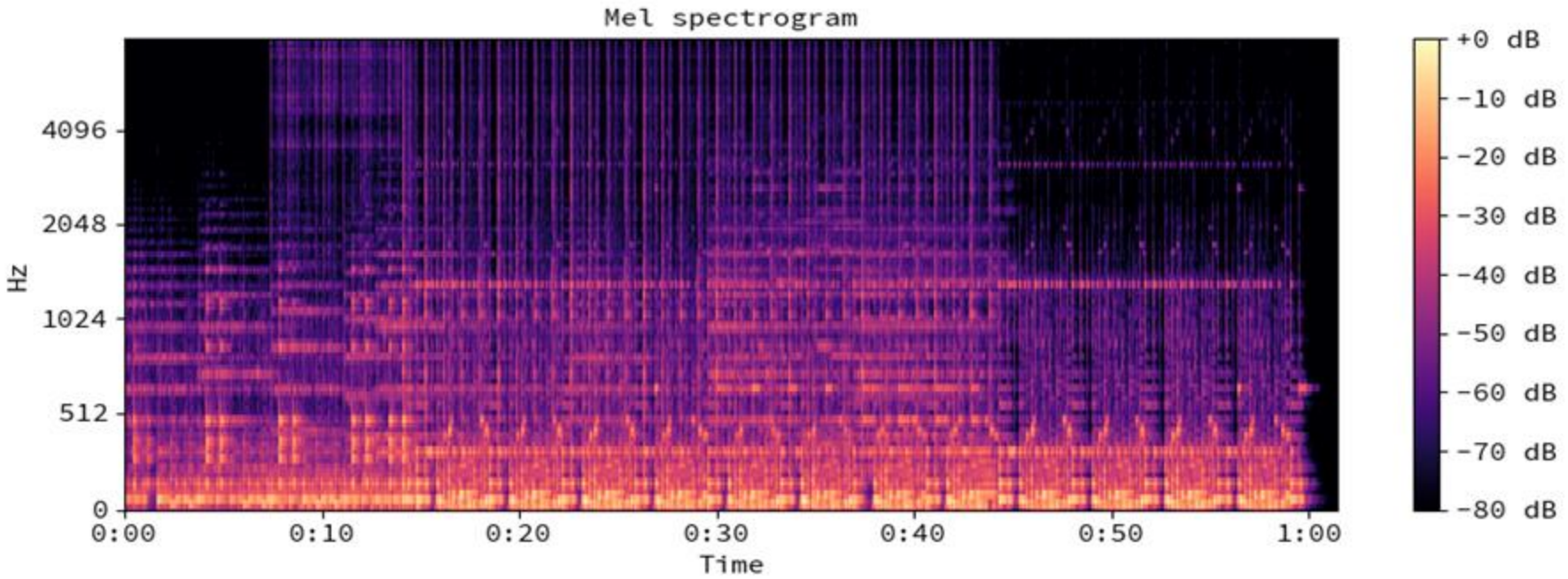




Performance on VoxCeleb

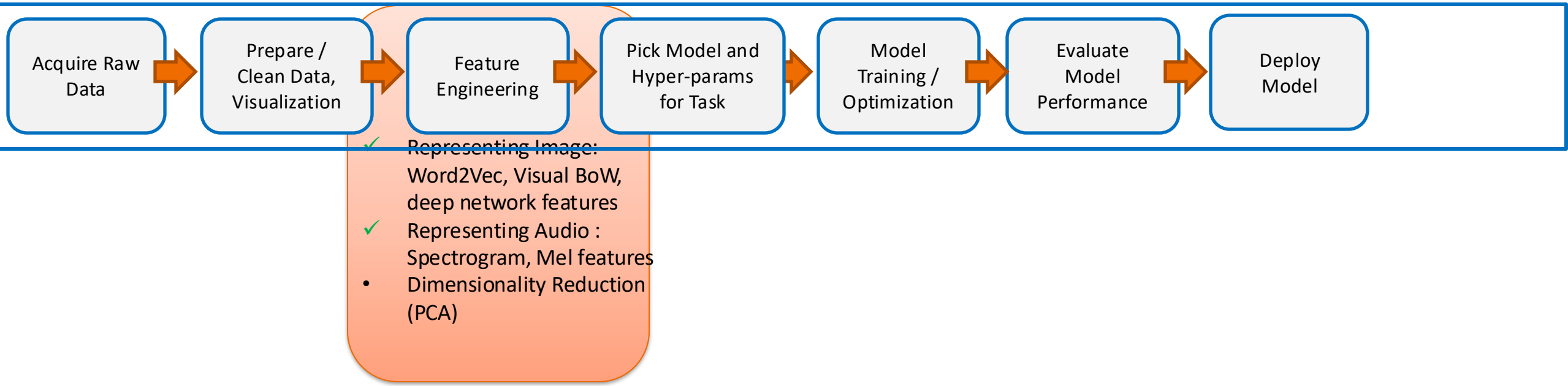
Accuracy	Top-1 (%)	Top-5 (%)
I-vectors + SVM	49.0	56.6
I-vectors + PLDA + SVM	60.8	75.6
CNN-fc-3s no var. norm.	63.5	80.3
CNN-fc-3s	72.4	87.4
CNN	**80.5**	**92.1**

Features from Mel Spectrogram



MFCC
(Hand coded Classic Features)

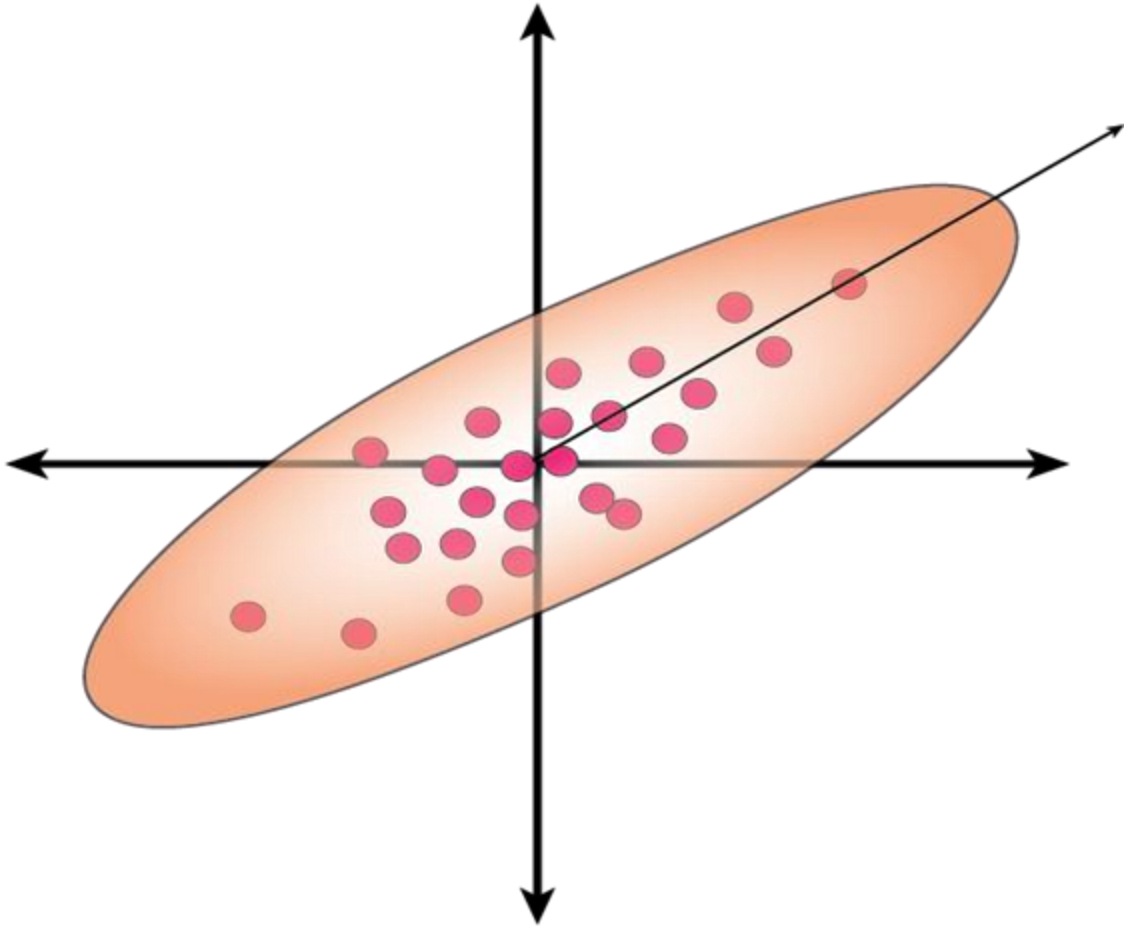
http://practicalcryptography.com/miscellaneous/machine-learning/guide-mel-frequency-cepstral-coefficients-mfccs/



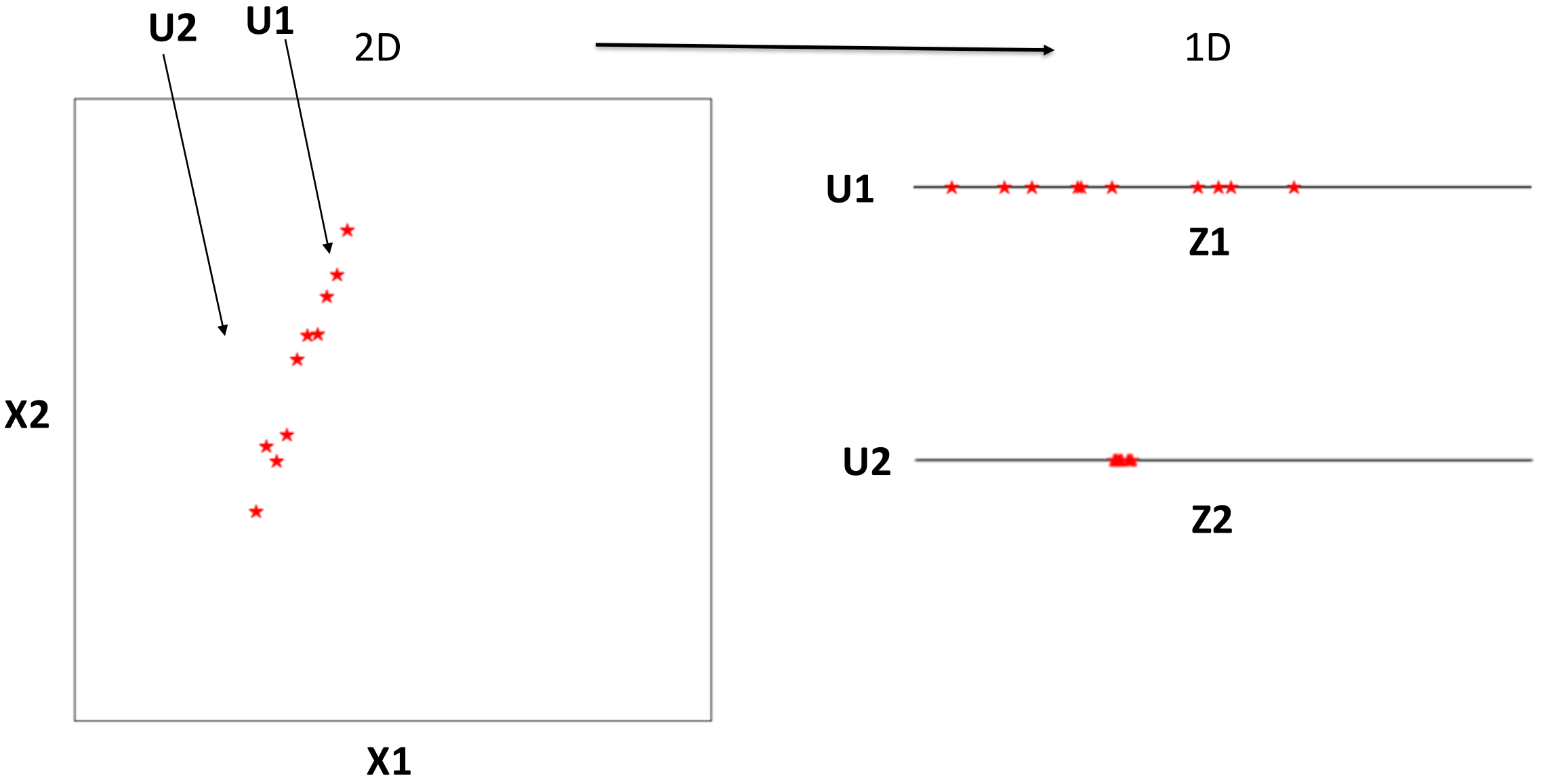
Principal Component Analysis

Simplifying Representations

PCA



Dimensionality and Representation



PCA

$$\widehat{\mu} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$\widehat{\sum} = \frac{1}{N}\sum_{i=1}^{N} (x_i - \widehat{\mu})(x_i - \widehat{\mu})^T$$

$$\begin{bmatrix} V_a & C_{a,b} & C_{a,c} & C_{a,d} & C_{a,e} \\ C_{a,b} & V_b & C_{b,c} & C_{b,d} & C_{b,e} \\ C_{a,c} & C_{b,c} & V_c & C_{c,d} & C_{c,e} \\ C_{a,d} & C_{b,d} & C_{c,d} & V_d & C_{d,e} \\ C_{a,e} & C_{b,e} & C_{c,e} & C_{d,e} & V_e \end{bmatrix}$$

Review:Eigenvector and Eigenvalue

$$Ax = \lambda x$$

A: Square Matrix

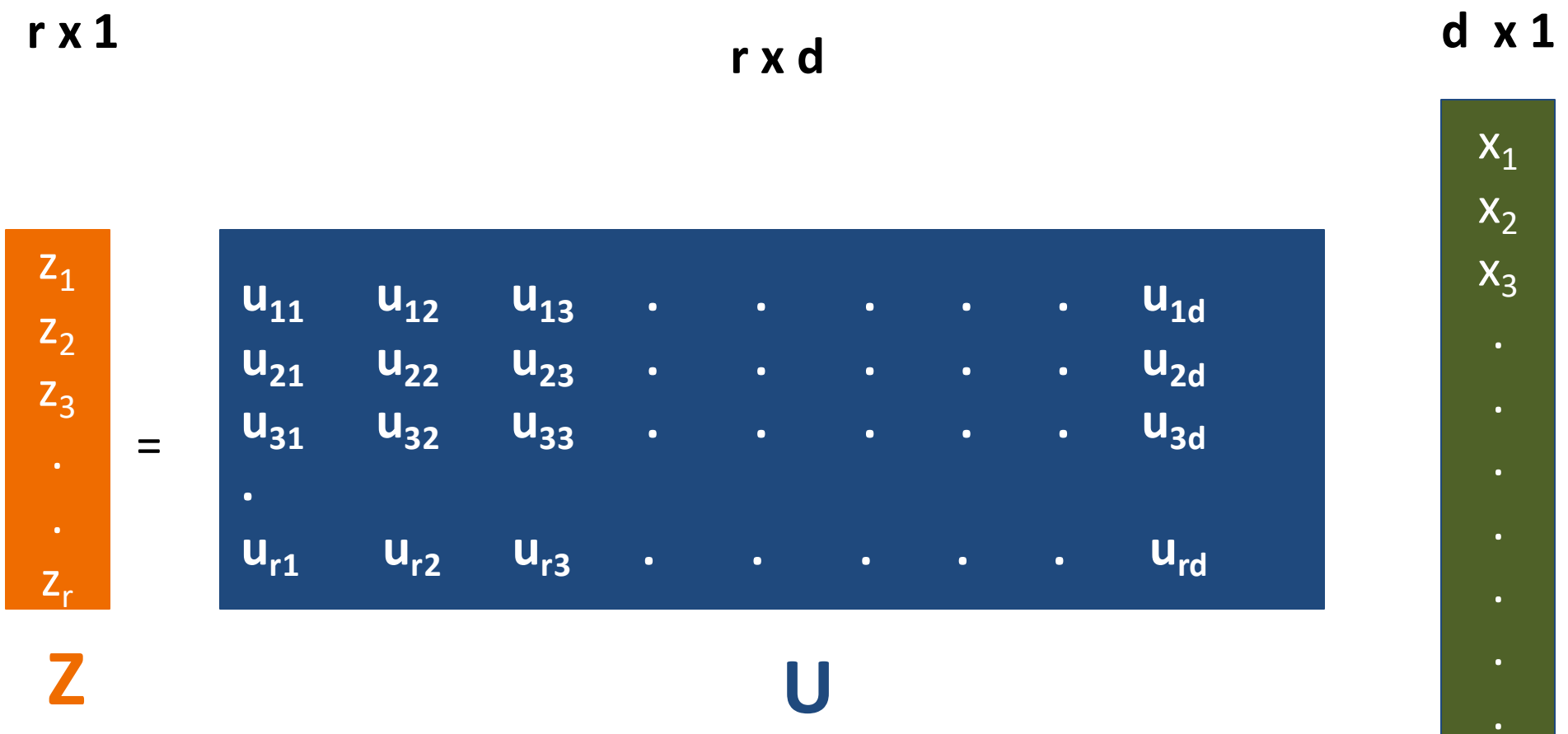
λ: Eigenvalue or characteristic value

x: Eigenvector or characteristic vector

- ***A is d X d***
- ***x is d x 1***
- ***Lambda is scalar***
- ***Max of d nonzero lambda***
- ***Min (N,d)***

$$\begin{bmatrix} 3 & 4 & -2 \\ 1 & 4 & -1 \\ 2 & 6 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 6 \end{bmatrix}$$

PCA based Feature Extraction



Each row in U is an Eigen vector of co-variance Matrix

Representing with EigenFaces

Any face in the database can be represented as a linear combination of the eigen faces.

PCA/Eigen Face Algorithm: Detail

1. Compute the mean feature vector

   $\mu = \frac{1}{p}\sum_{k=1}^{p} x_k$, where, x_k is a pattern ($k = 1$ to p), p = number of patterns, x is the feature matrix

2. Find the covariance matrix

   $C = \frac{1}{p}\sum_{k=1}^{p}\{x_k - \mu\}\{x_k - \mu\}^T$ where, T represents matrix transposition

3. Compute Eigen values λ_i and Eigen vectors v_i of covariance matrix

   $Cv_i = \lambda_i v_i$ $(i = 1, 2, 3,....q)$, q = number of features

4. Estimating high-valued Eigen vectors

   (i) Arrange all the Eigen values (λ_i) in descending order

   (ii) Choose a threshold value, θ

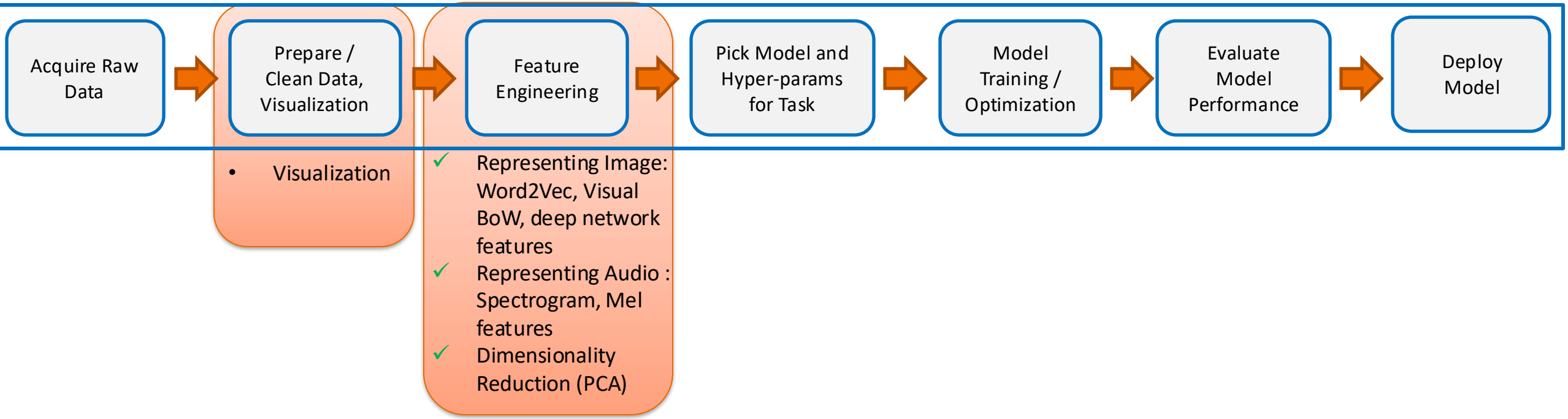
   (iii) Number of high-valued λ_i can be chosen so as to satisfy the relationship

   $$\left(\sum_{i=1}^{s}\lambda_i\right)\left(\sum_{i=1}^{q}\lambda_i\right)^{-1} \geq \theta$$, where, s = number of high valued λ_i chosen

   (iv) Select Eigen vectors corresponding to selected high valued λ_i

5. Extract low dimensional feature vectors (principal components) from raw feature matrix.

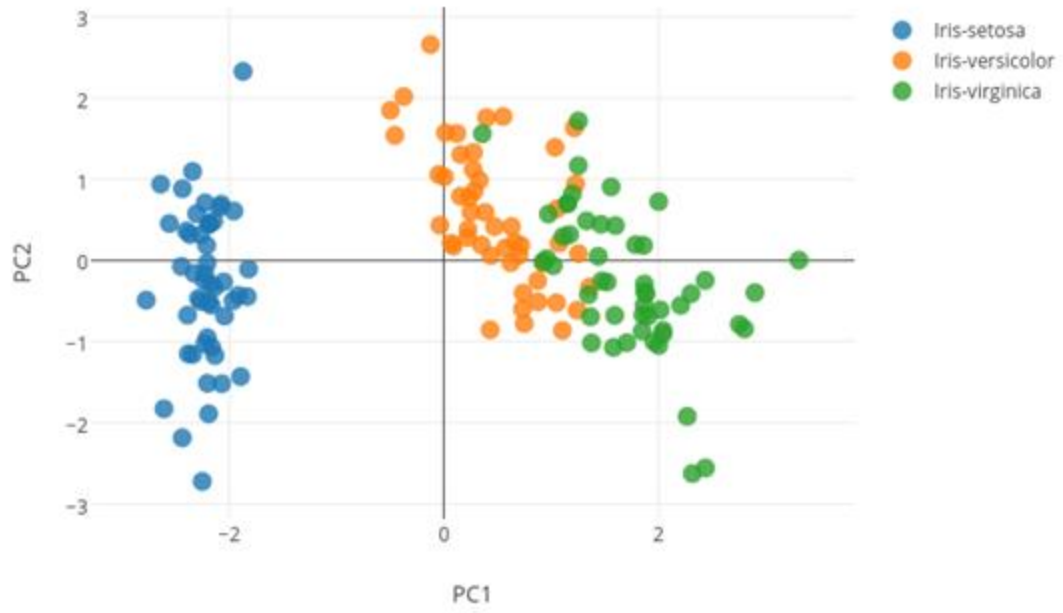
   $P = V^T x$, where, V is the matrix of principal components and x is the feature matrix



Data Visualization

When Data is High-Dimensional

PCA



PLOT ON 2 PRINCIPLE COMPONENTS

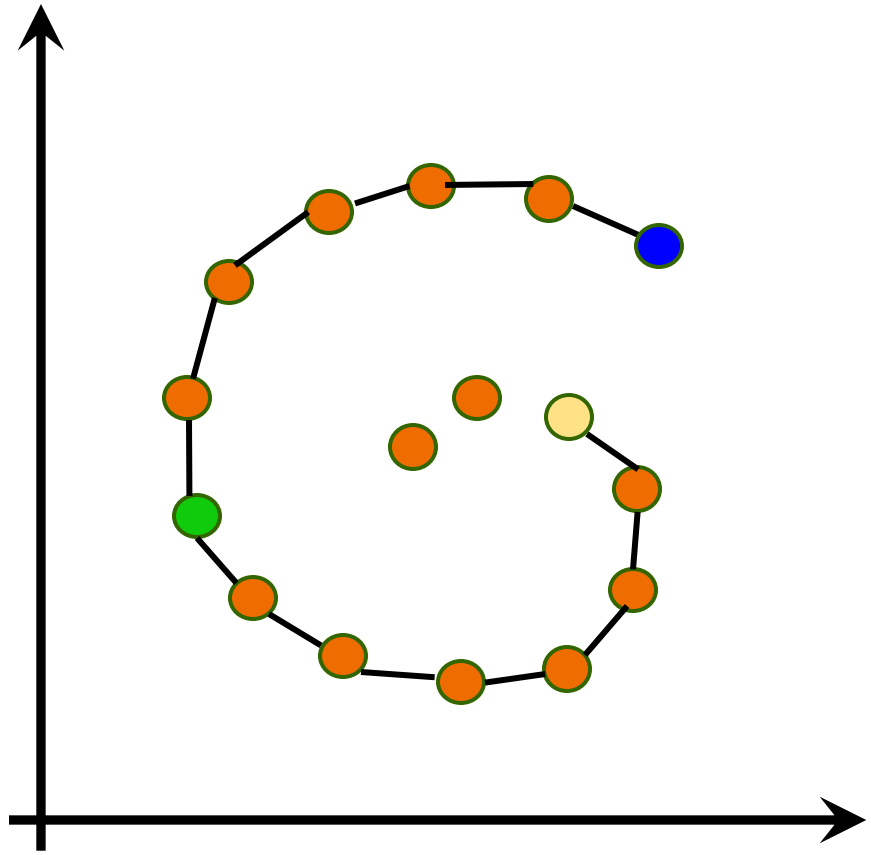
MDS (Multidimensional scaling)

- Minimize an objective function that measures the discrepancy between similarities in the data and similarities in the map.

- Distance between samples in “high” dimension and “low’’ dimension is same (or D-d) is minimized.

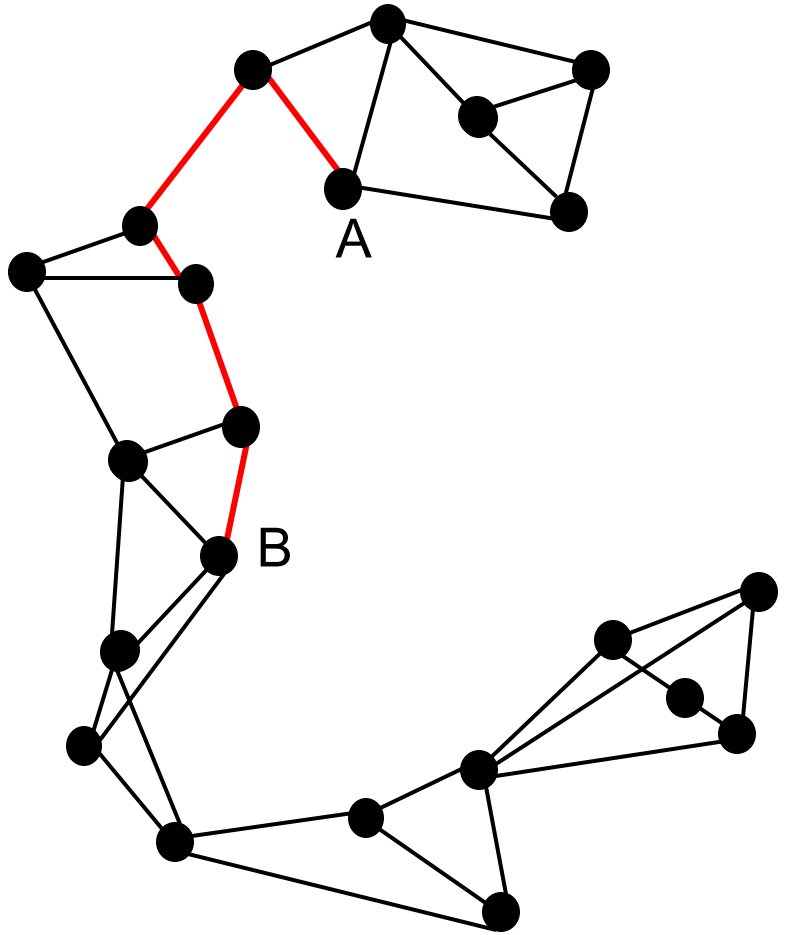
ISOMAP (Isometric Mapping)

- d(●,○) > d(●,●)
- Is Euclidean metric the right distance metric?
- How to robustly measure distances along the manifold?



ISOMAP

- How does ISOMAP measure the MD?
- Connect each data point to its K nearest neighbors in the high-dimensional space.
- Link weights: True Euclidean distances.
- $MD(A, B) = ShortestPath(A, B)$ in this ***neighborhood graph***.
- Compute the low-dimensional embedding as in Metric MDS.



LLE: Locally Linear Embedding

- Idea: Preserve the structure of local neighbourhood

$$\mathbf{x}_i \approx \sum_j w_{ij}\mathbf{x}_j$$

- Approach:
  - —Represent each point as a weighted combination of its Neighbours in HD. Remember the $w_{ij}s$.
  - —Find a LD representation that minimize the representation error:

$$Cost = \sum_i \| \mathbf{y}_i - \sum_{j \varepsilon\, N(i)} w_{ij}\mathbf{y}_j \|^2$$

  - The weights w_{ij} refer to the amount of contribution the point x_i has while reconstructing the point x_i. The cost function is minimized under two constraints: (a) Each data point x_i is reconstructed only from its neighbors, thus enforcing w_{ij} to be zero if point x_j is not a neighbor of the point x_i and (b) The sum of every row of the weight matrix equals 1.
  - Also ys should have unit variance across each dimension.

SNE and t-SNE

- Idea is simple: Instead of distance think about probabilities. P_{ij} as the probability of j in the neighborhood of i.
- For each point, we have now a probability vector (of size N).
  - SNE uses Gaussian. T-SNE uses another t-distribution (with 1 degree of freedom).
- We want these prob vectors to be the same in low dimensional.
- Optimize using gradient descent.

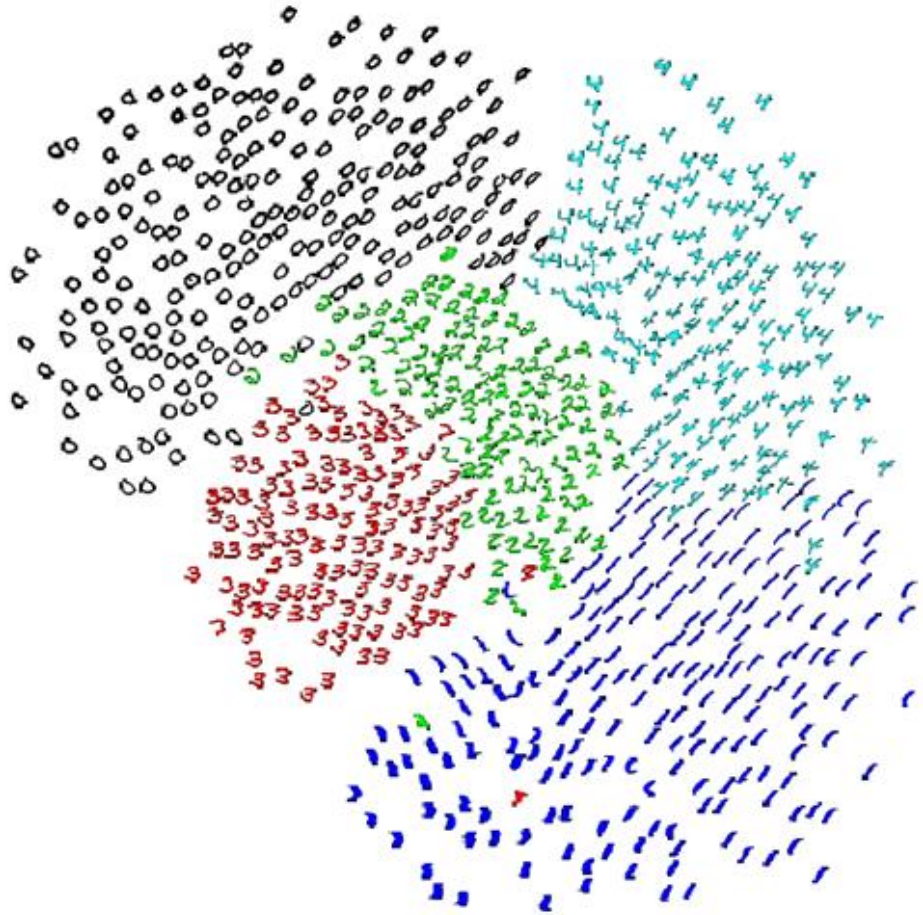
Computing the LD Embedding

$$Cost = \sum_i KL\left(P_j \parallel Q_i\right) = \sum_i \sum_j p_{j|i} \log \frac{p_{j|i}}{q_{j|i}}$$

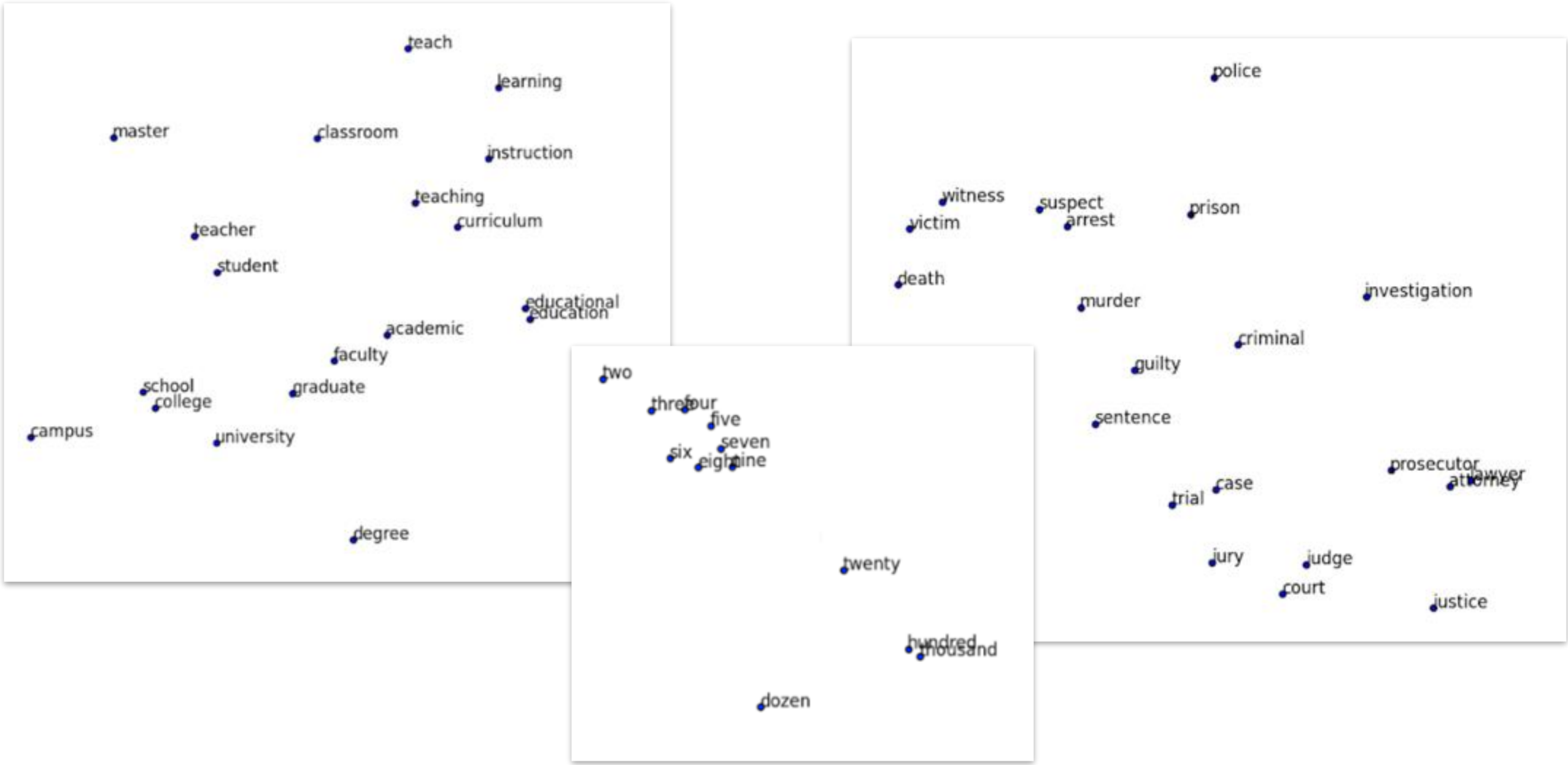
- For points where p_{ij} is large and q_{ij} is small we lose a lot.
  - Nearby points in high-D really want to be nearby in low-D

PCA on MNIST *(0-9)*

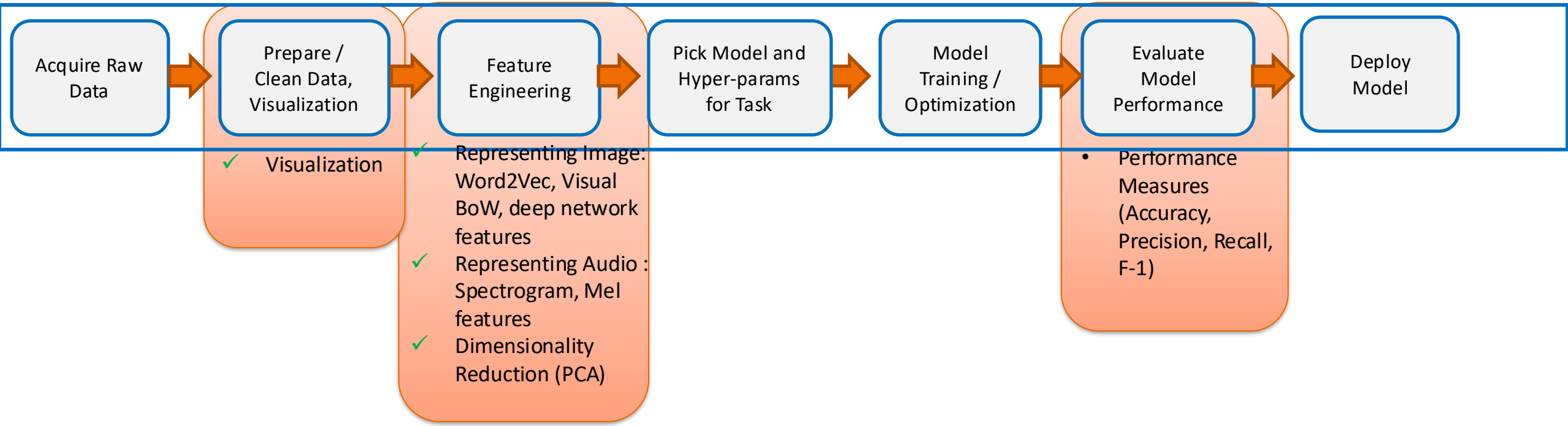
SNE on MNIST *(0-5)*



Word2vec



http://nlp.yvespeirsman.be/blog/visualizing-word-embeddings-with-tsne/



Performance Metrics

Key accuracy measures and terminologies

- Classification Error = $\frac{errors}{total}$

$$= \frac{FP + FN}{TP + TN + FP + FN}$$

n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

- Accuracy = 1 - Error = $\frac{correct}{total}$

$$= \frac{TP + TN}{TP + TN + FP + FN}$$

Revisiting scenarios where metrics are appropriate

- When you do cancer screening what do you care?
  - High TP and Low FN
- When you classify between “apple” and “orange”
  - High Accuracy
- Automatic Firing on detecting a violation.
  - Very low FP

Precision and Recall

$$Precision = \frac{TP}{(TP + FP)}$$

$$Recall = \frac{TP}{(TP + FN)}$$

Precision and Recall – examples

- A system which needs to launch a missile at a terrorist hideout located in a dense urban area.
- Precision not 100% ➡ civilian casualties
- A system which needs to identify cancer-risk patients
- Recall not 100% ➡ some patients will die of cancer

F-measure: Combines Precision and Recall

- What to do when one classifier has better Precision but worse Recall, while other classifier behaves exactly opposite?
  - F-measure (Information Retrieval)

$$F_1 = \frac{2}{\frac{1}{Recall} + \frac{1}{Precision}}$$

F-measure

$$Precision = \frac{TP}{(TP + FP)} \qquad Recall = \frac{TP}{(TP + FN)}$$

- What to do when one classifier has better Precision but worse Recall, while other classifier behaves exactly opposite?
  - F-measure (Information Retrieval)

$$F_1 = \frac{2}{\frac{1}{Recall} + \frac{1}{Precision}}$$

- F1 measure punishes extreme values more !
- Definition of Recall and Precision have same numerator, different denominators. A sensible way to combine them is harmonic mean.

F-measure

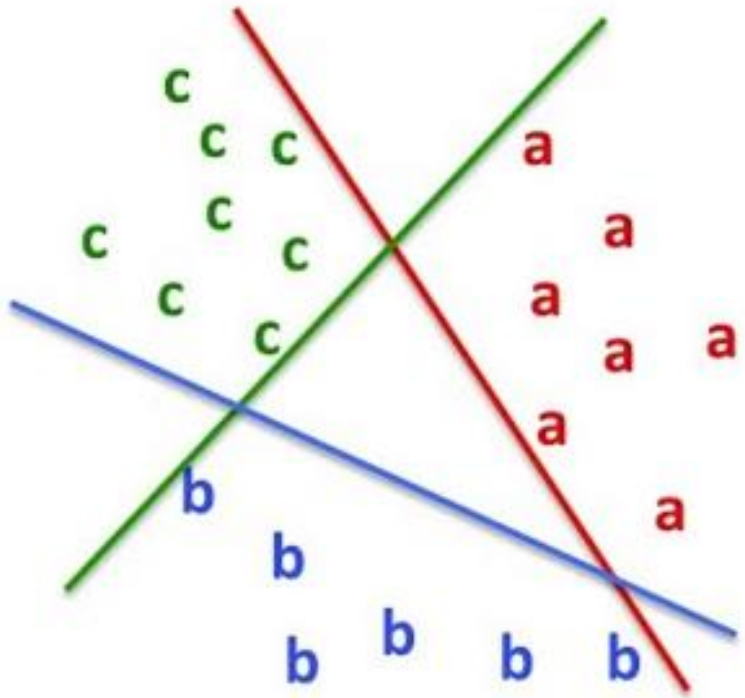
- Use when
  - FP and FN are ‘equally costly’
  - You don’t expect results to change when more data is added
  - TN is high (e.g. face detector)

$$Precision = \frac{TP}{(TP + FP)} \qquad Recall = \frac{TP}{(TP + FN)}$$

$$\frac{2}{\frac{1}{Recall} + \frac{1}{Precision}}$$

How to use 2-class measures for multi-class ?

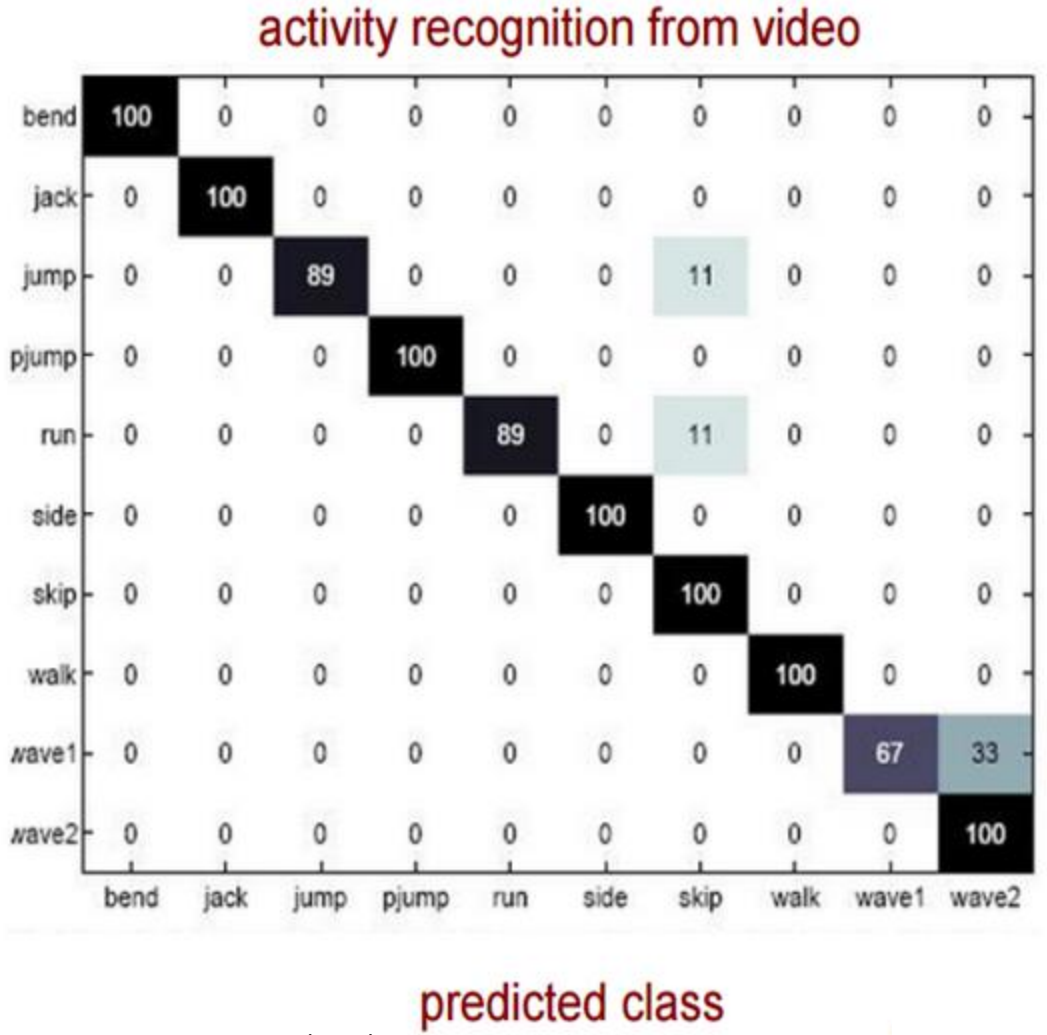
- Convert into 2-class problem(s) !



Multi-class problems - Confusion matrix

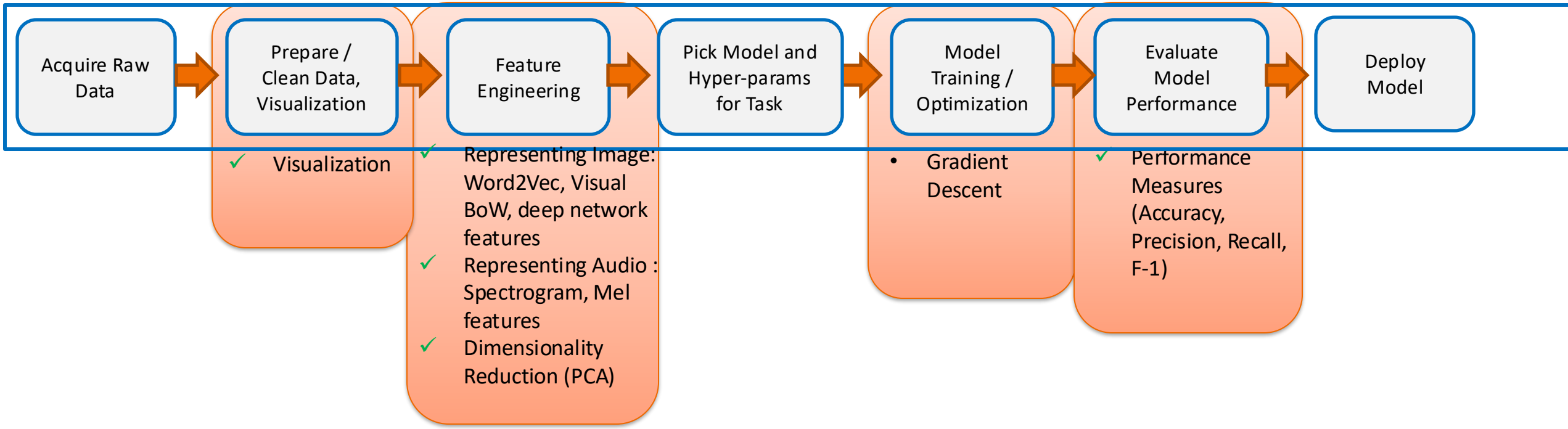
n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

actual class



Courtesy: vision.jhu.edu

Avg. accuracy may not be very meaningful with imbalanced class label distribution



Gradient Descent

The Problem

- Find $\boldsymbol{w}$, given examples: $(x_i, y_i), i = 1,2, \ldots, n$
- Supervised situation: output label y; is available for all training n samples
- **Objective:** predict y values for a novel input x that we have not seen before
  - Called generalization in Machine Learning
- How do we find w? **Ans: Gradient descent!**

Loss Function

- **Error/Loss (L):** A function of the difference between the actual value or label (y_i) and the predicted value ($f(w, x_i)$)

- **Total Loss:**

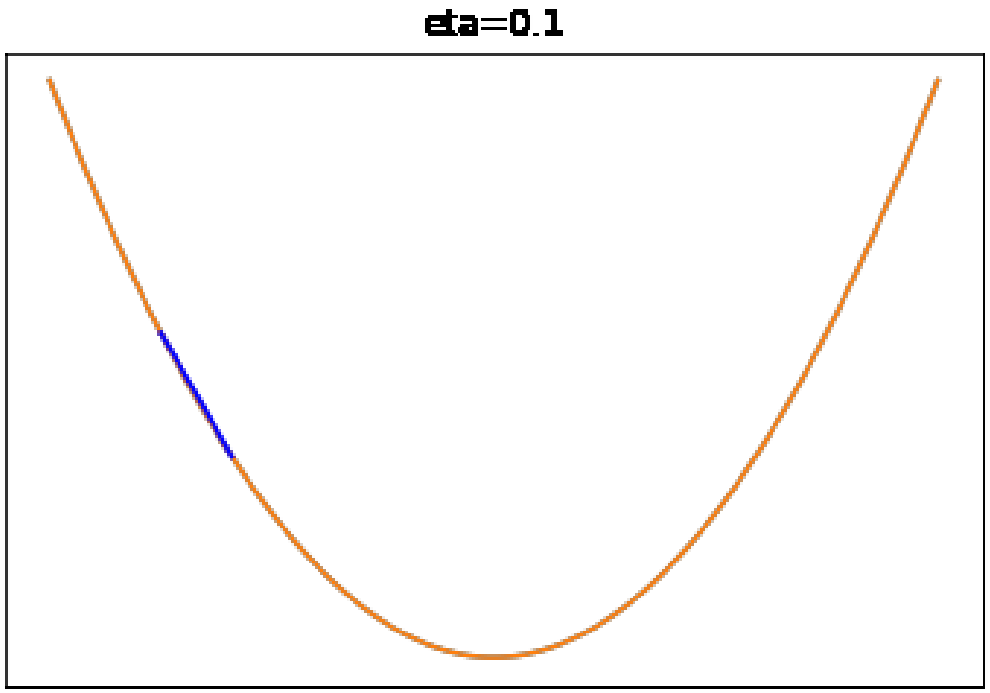
$$J = \sum_{i=1}^{n} L(y_i, f(w, x_i))$$

the sum of loss over n labelled training samples: (x_i, y_i)

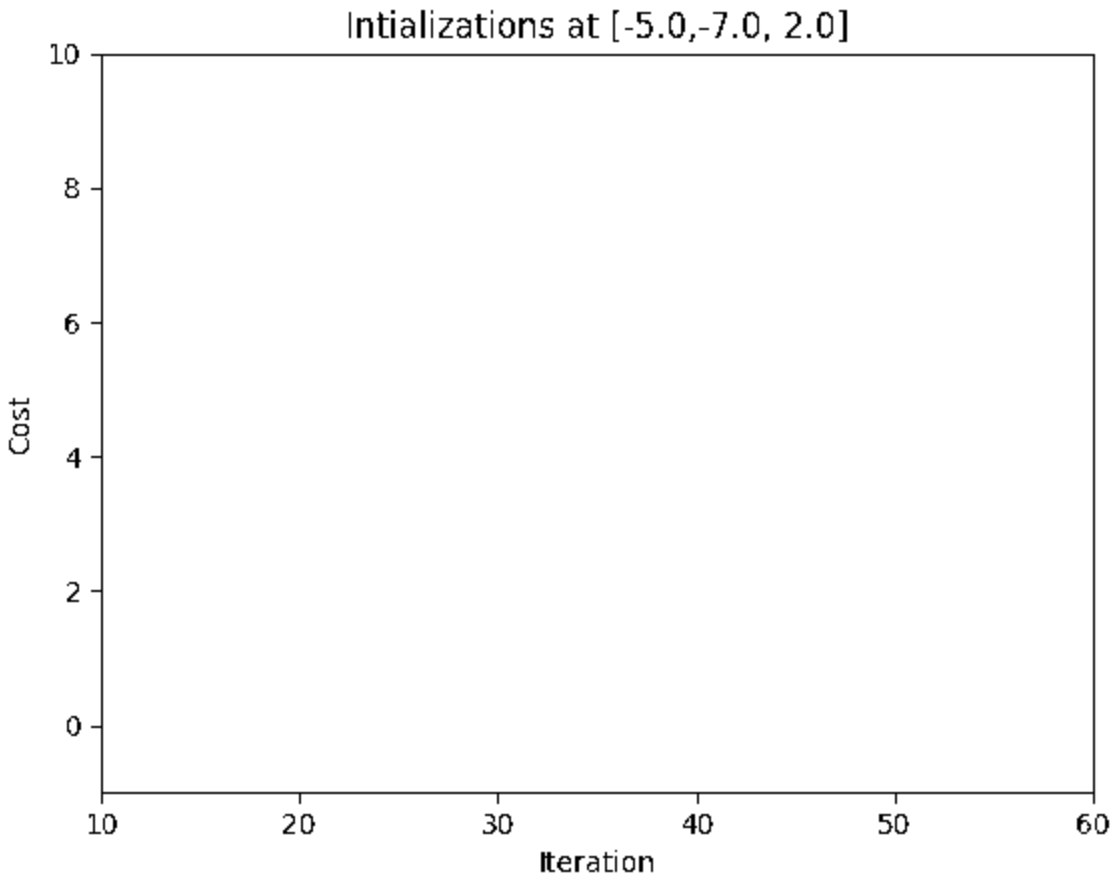
- **Strategy:** Start with some initial values for w and bring the predicted values closer to the corresponding labels (or minimize J) by adjusting w.

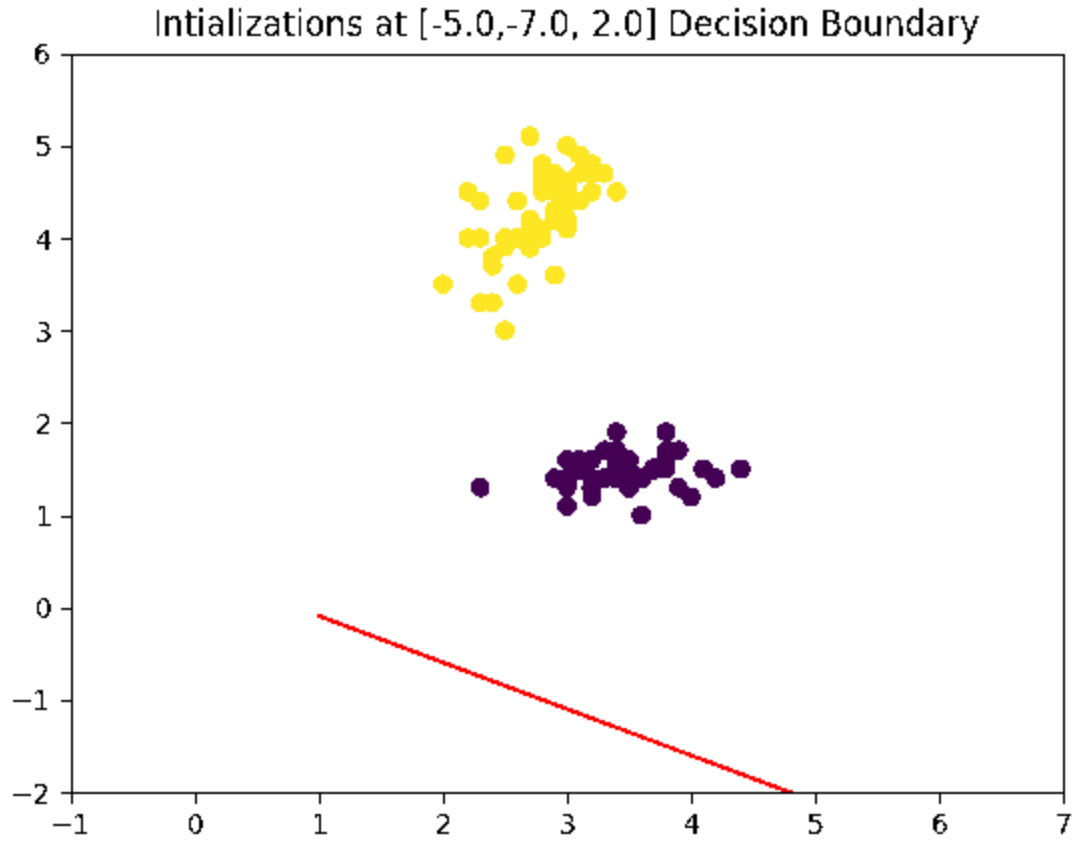
Gradient Descent

- Minimum of the function lies in the opposite direction of gradient
- Start with a guess
- Take a step against gradient:
- $\boldsymbol{w' = w - \eta \nabla J(w)}$

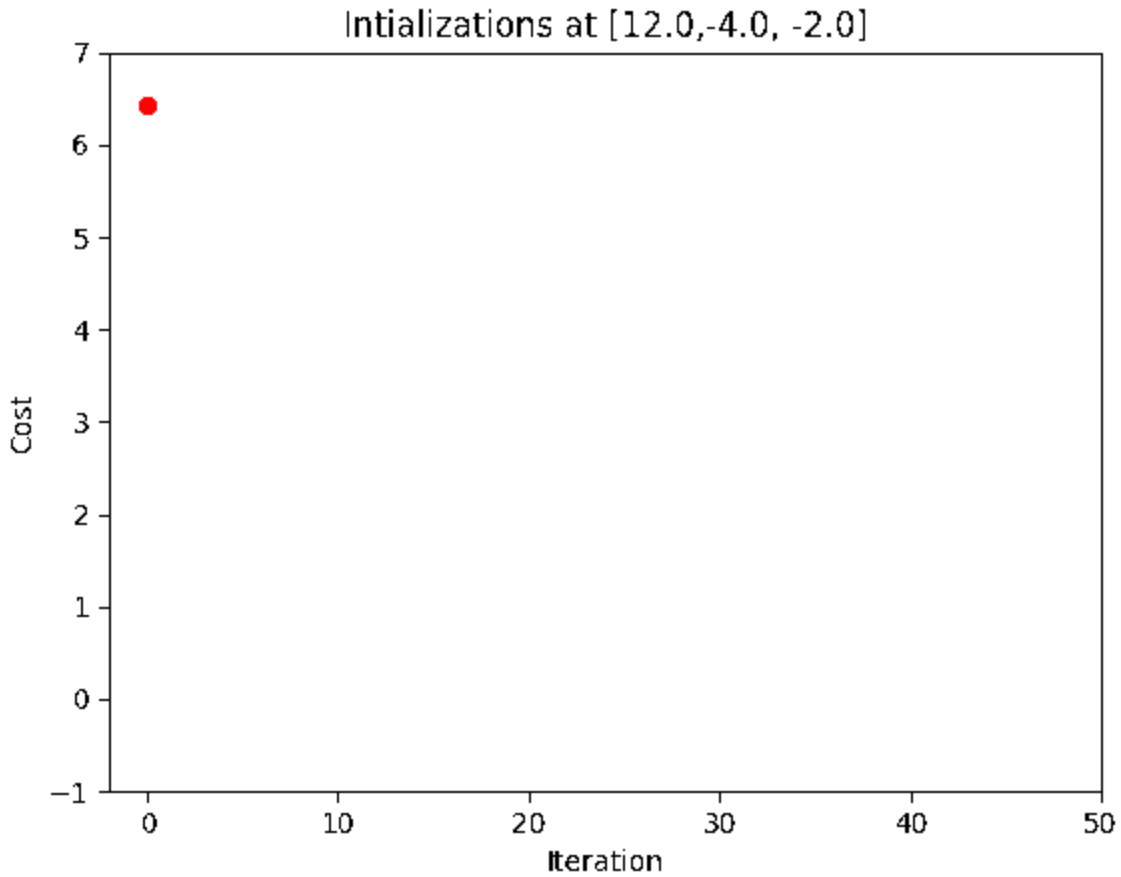


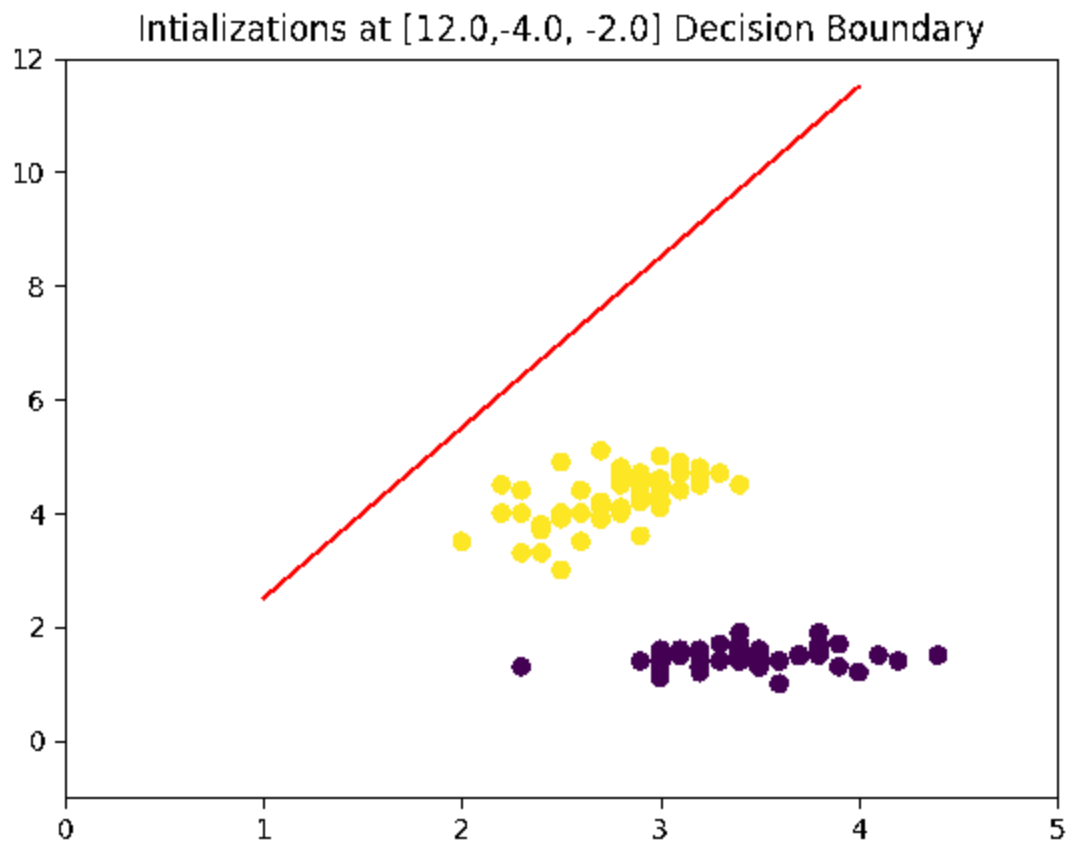
Gradient Descent - Initialization





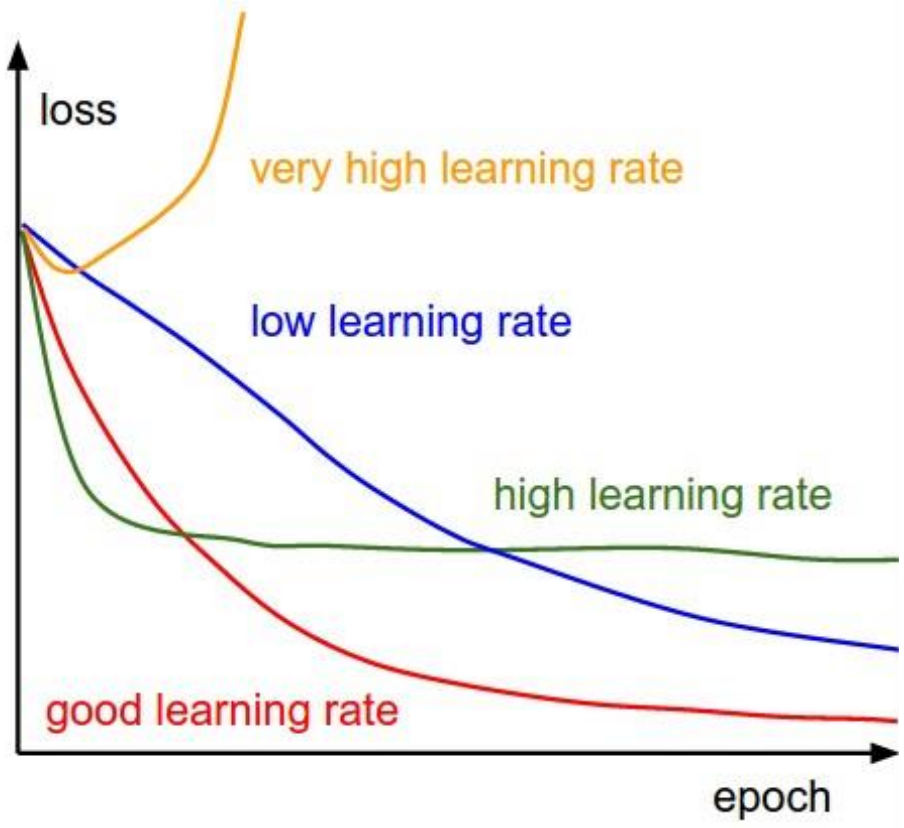
Gradient Descent - Initialization





Scale varied for better visibility

Gradient Descent - Learning rate

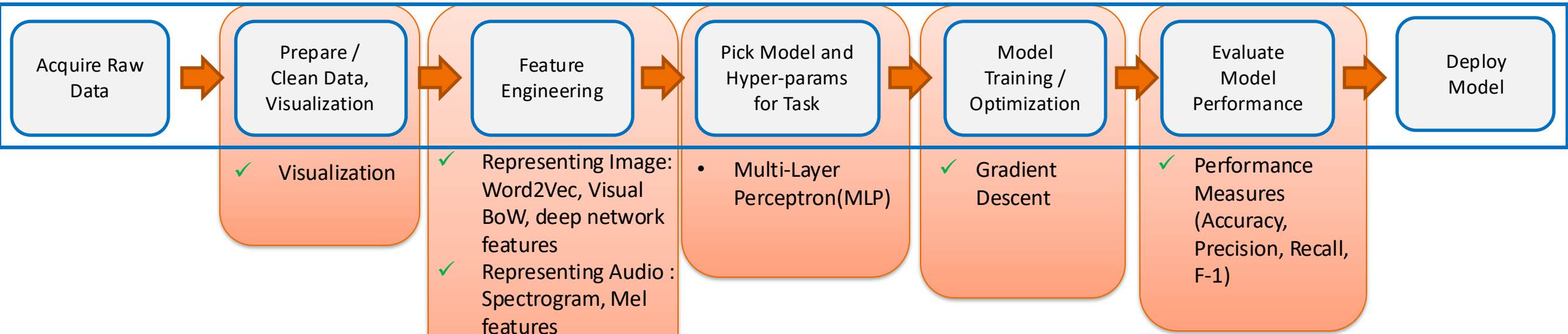


http://cs231n.github.io/neural-networks-3/#baby

Mini Batch Gradient Descent

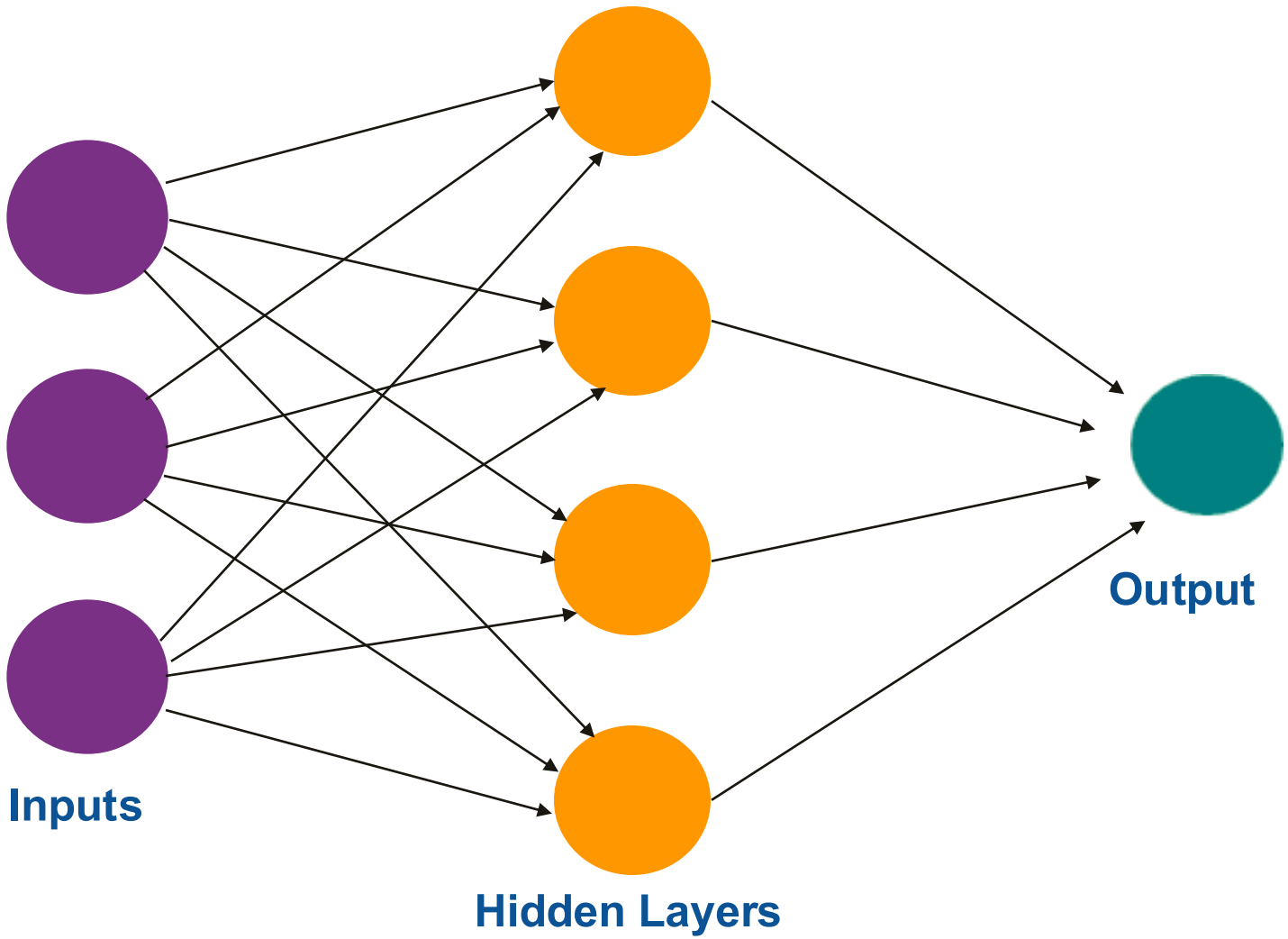
Pseudo Code

```
model = initialization(...)
train_data = load_training_data()
Tb1, Tb2, Tb3, ……,Tbm = split_training_batches(train_data)
for i in 1...n:
        error = 0
        for Tb in Tb1, Tb2, Tb3, ……,Tbn :
                for X, y in Tb :
                        predictions = predict(X, model)
                        error += calculate_error(y,predictions)
                gradient  = differentiate(error)
                model = update_model(model, gradient)
```


Acquire Raw Data
Prepare / Clean Data, Visualization
Feature Engineering
Pick Model and Hyper-params for Task
Model Training / Optimization
Evaluate Model Performance
Deploy Model
✓ Visualization
✓ Representing Image: Word2Vec, Visual BoW, deep network features
✓ Representing Audio : Spectrogram, Mel features
✓ Dimensionality Reduction (PCA)
• Multi-Layer Perceptron(MLP)
✓ Gradient Descent
✓ Performance Measures (Accuracy, Precision, Recall, F-1)

MLP

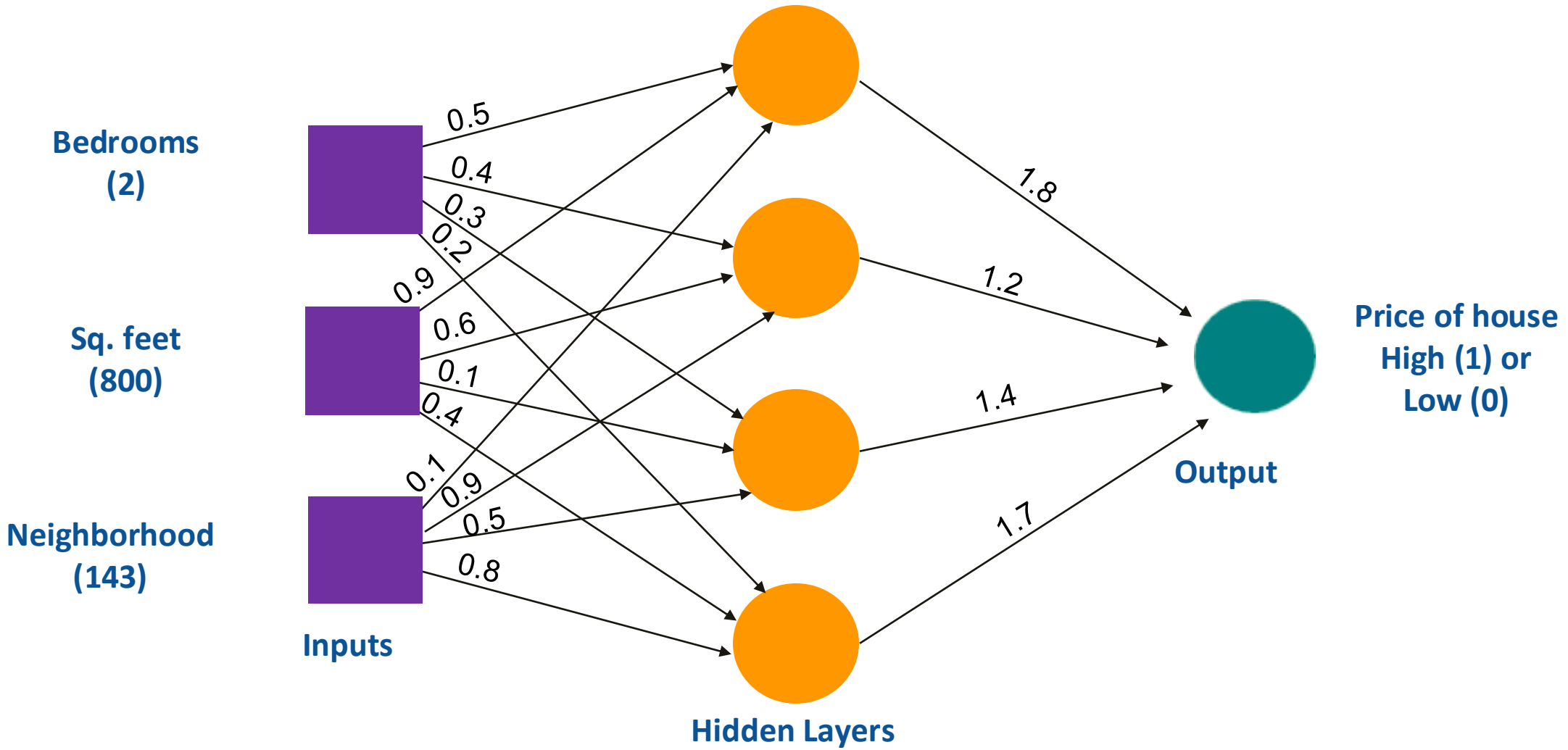
MLP



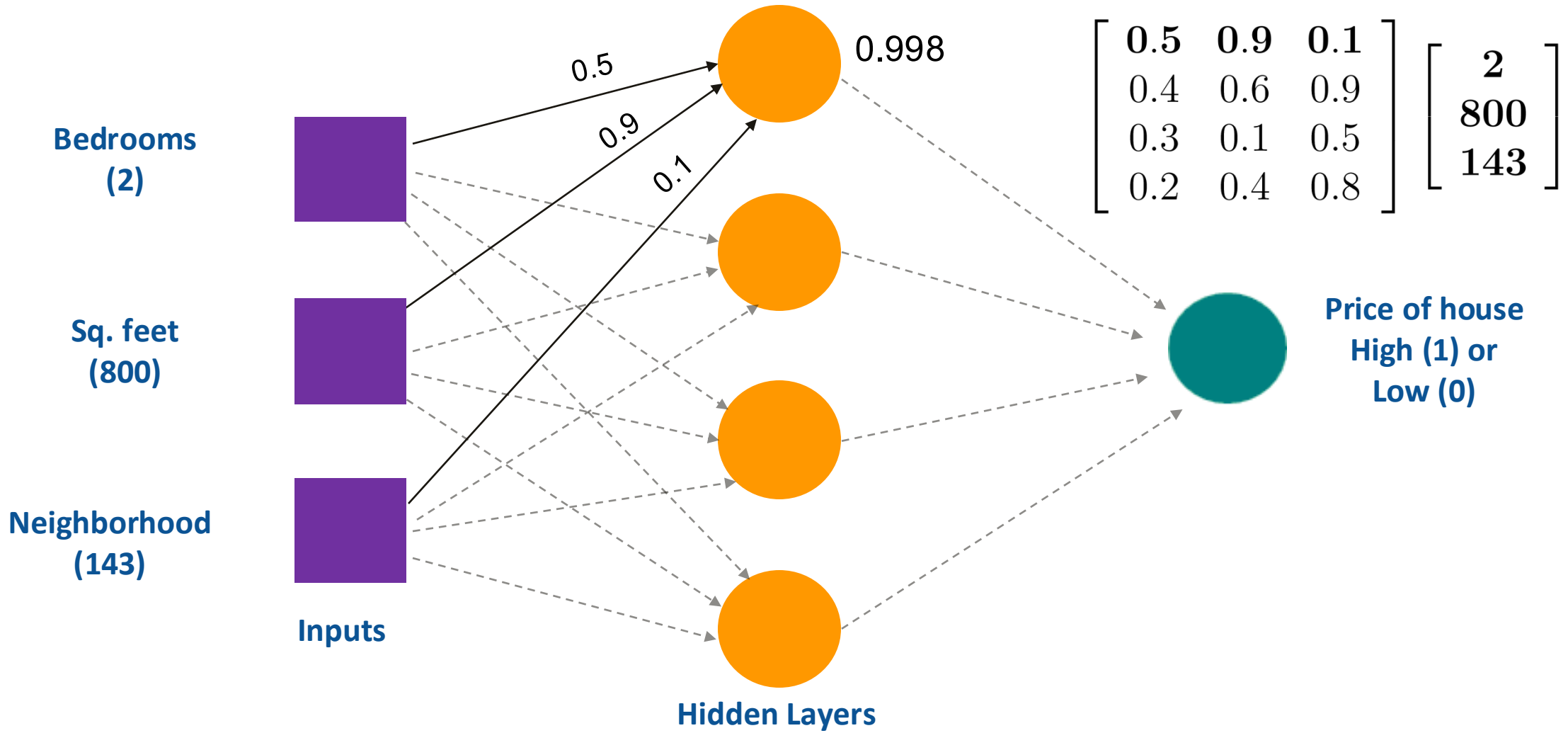
Eg. House price from attributes

Bedrooms	Sq. Feet	Neighborhood (no. of houses in the locality)	Price high or low? High (1), Low (0)
3	2000	90	1
2	800	143	0
2	850	167	0
1	550	267	0
4	2000	396	1

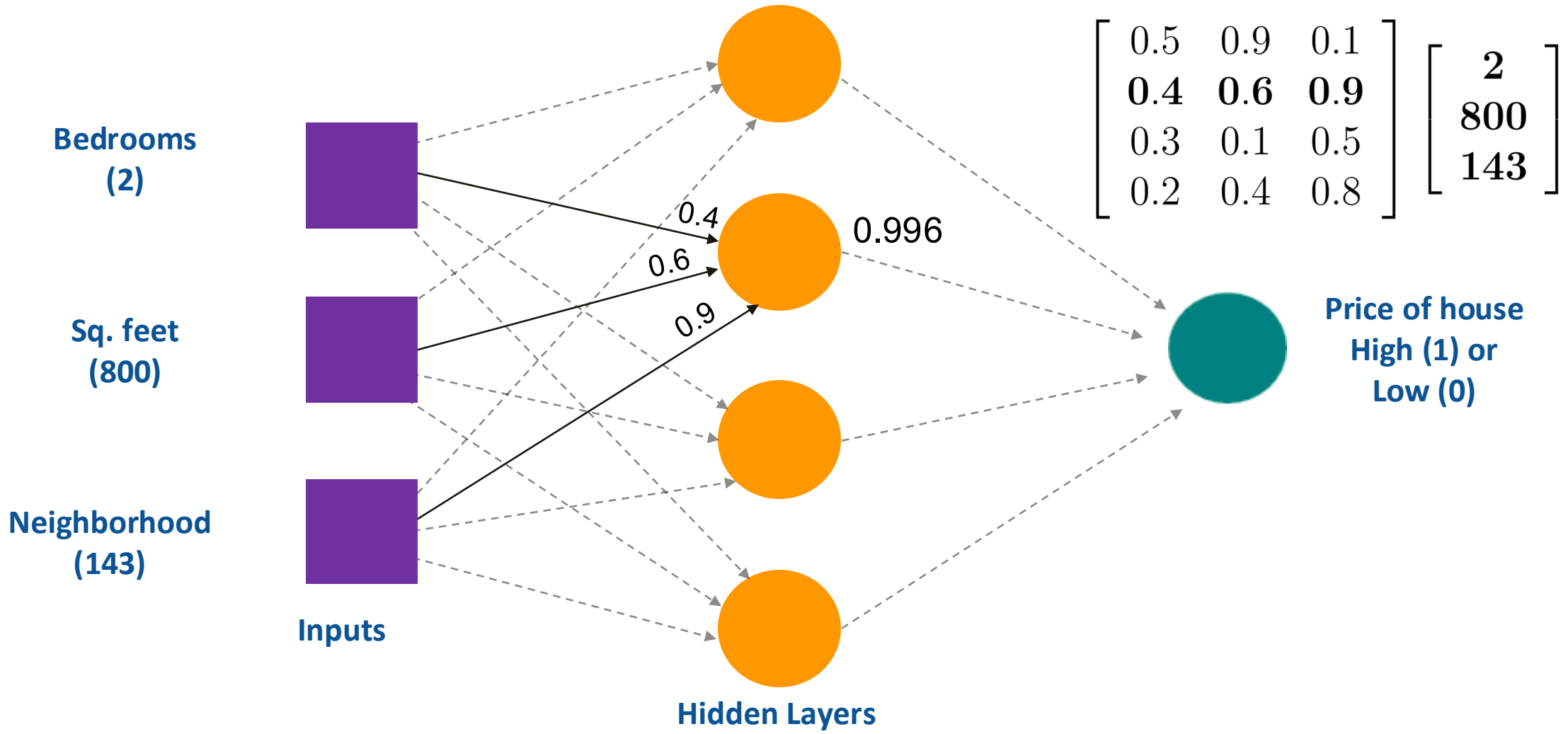
Initialize weights



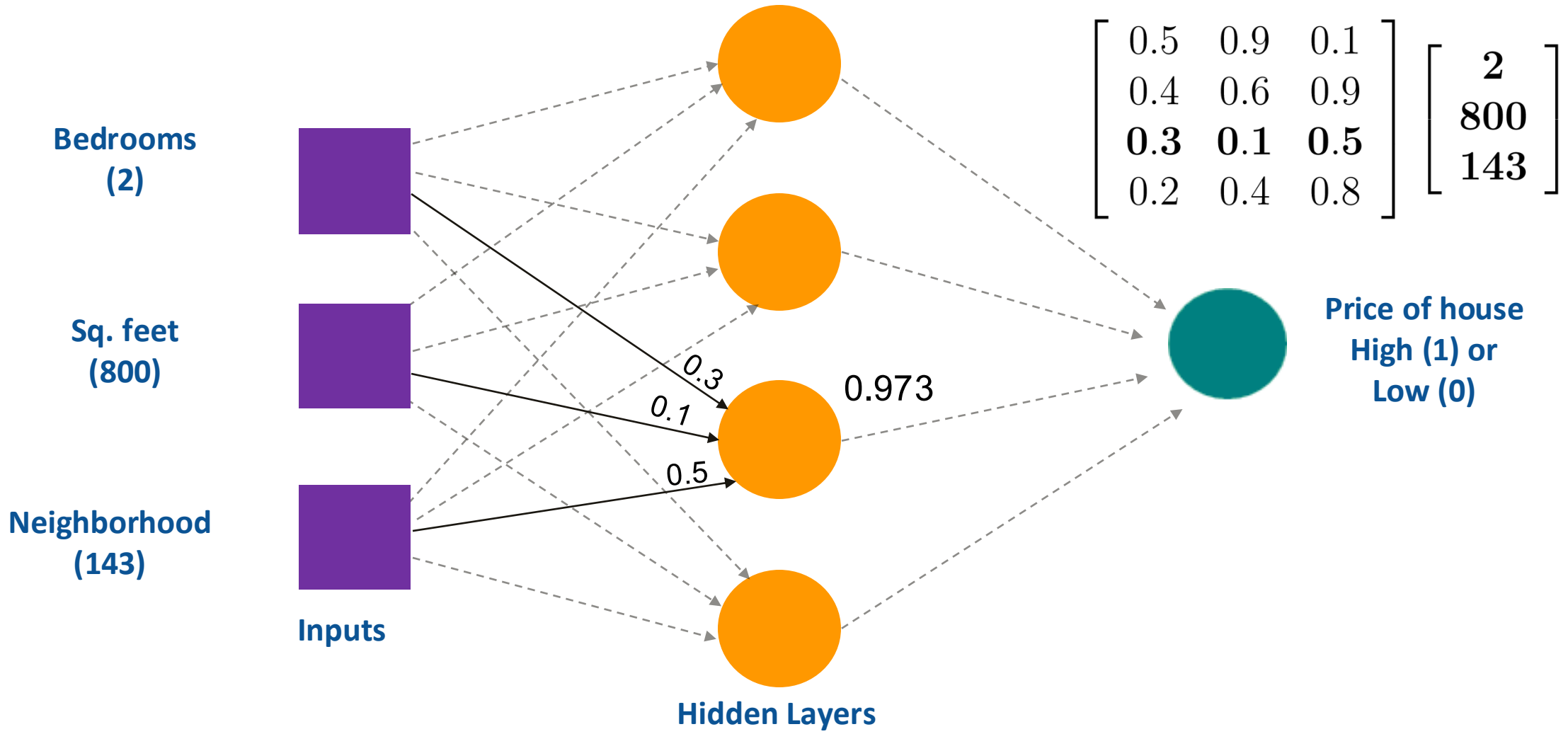
Weights at the first neuron



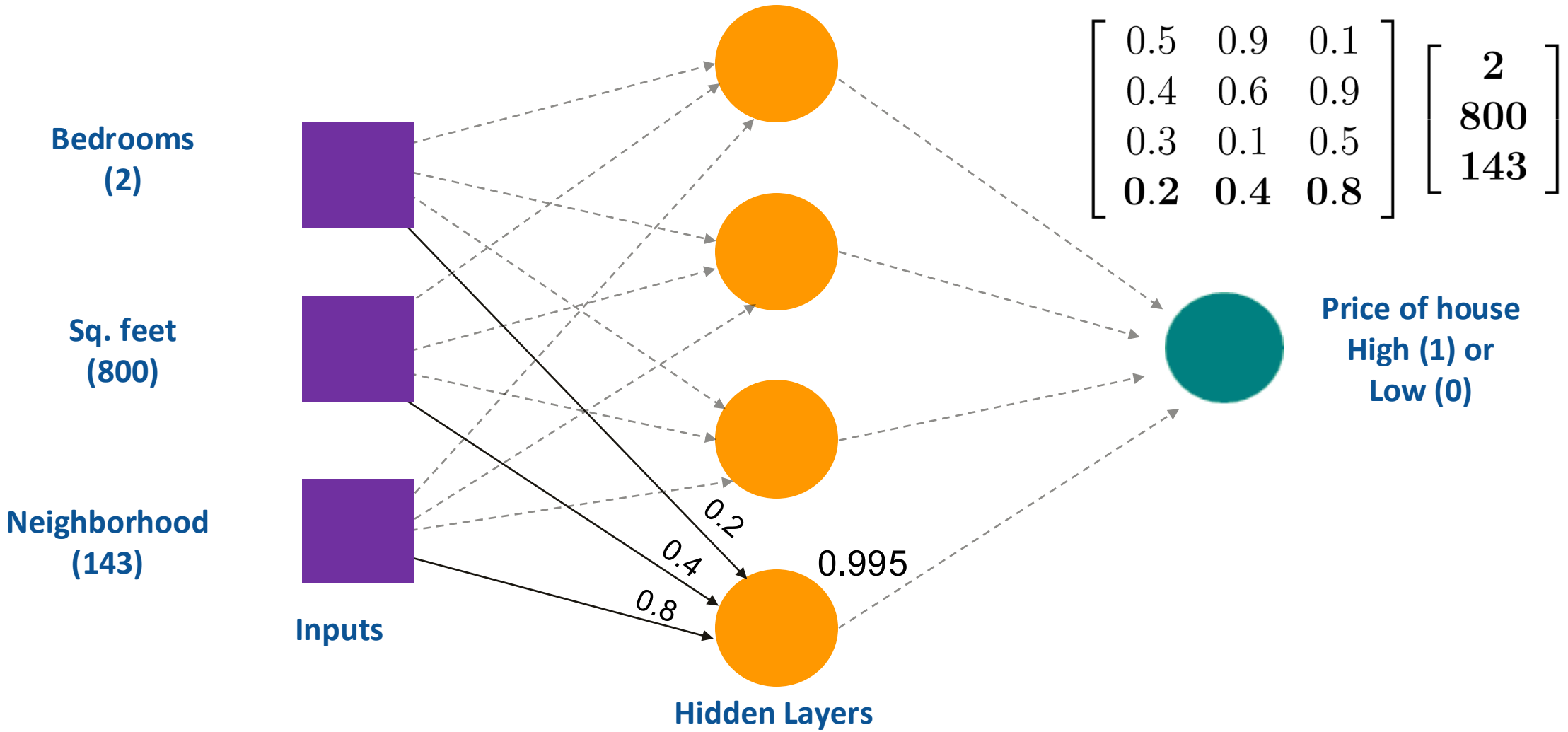
Weights at the second neuron



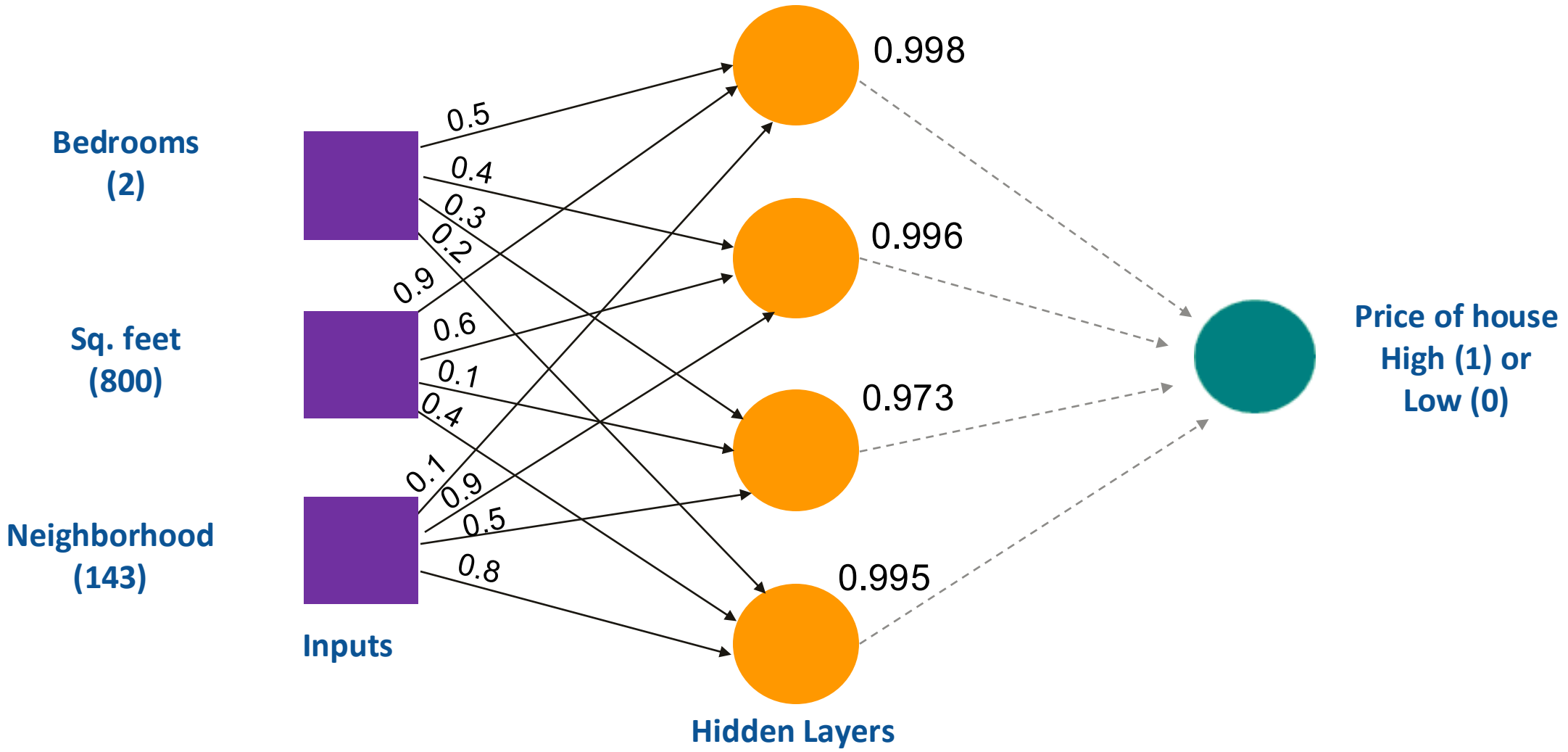
Weights at the third neuron



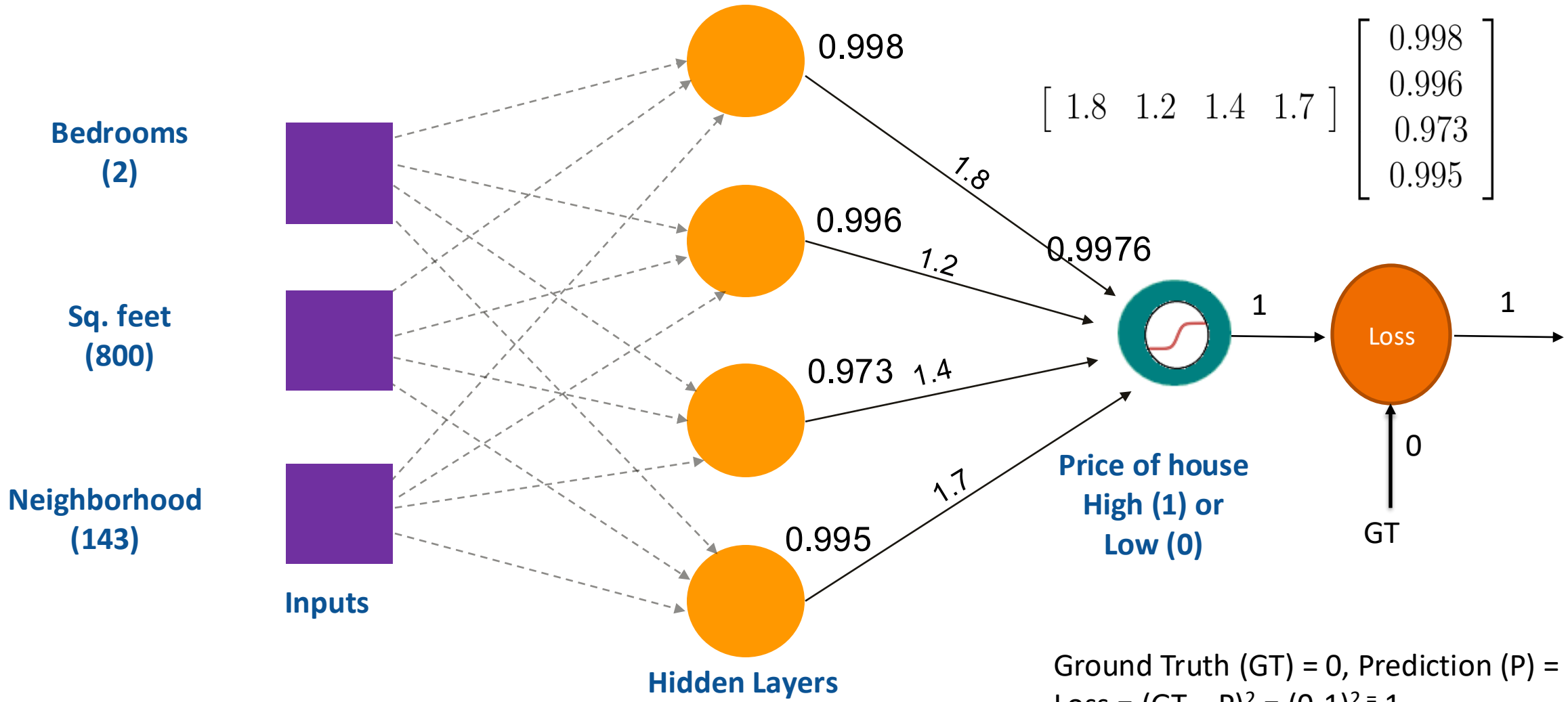
Weights at the fourth neuron



Inputs to the next layer



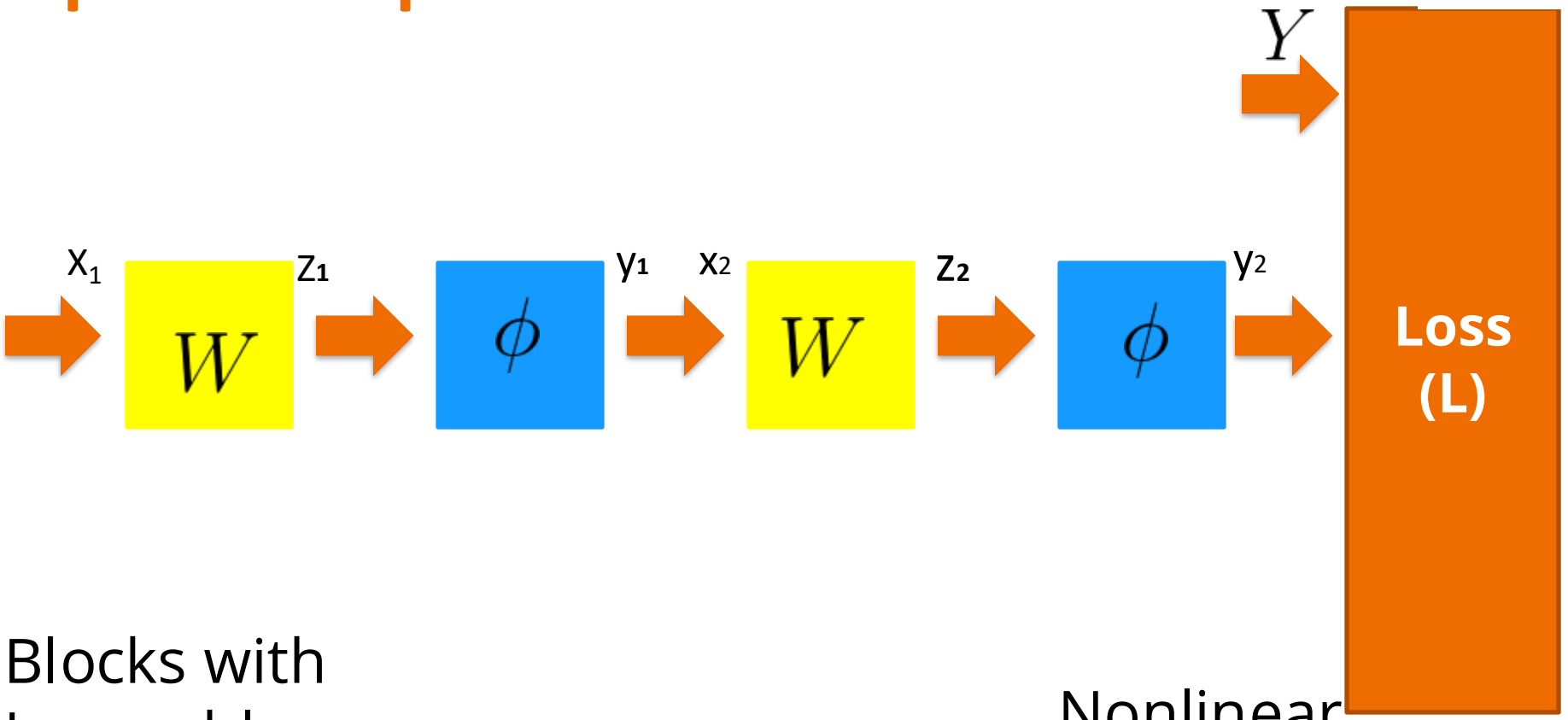
Computations in next layer



Ground Truth (GT) = 0, Prediction (P) = 1

Loss = $(GT - P)^2 = (0-1)^2 = 1$

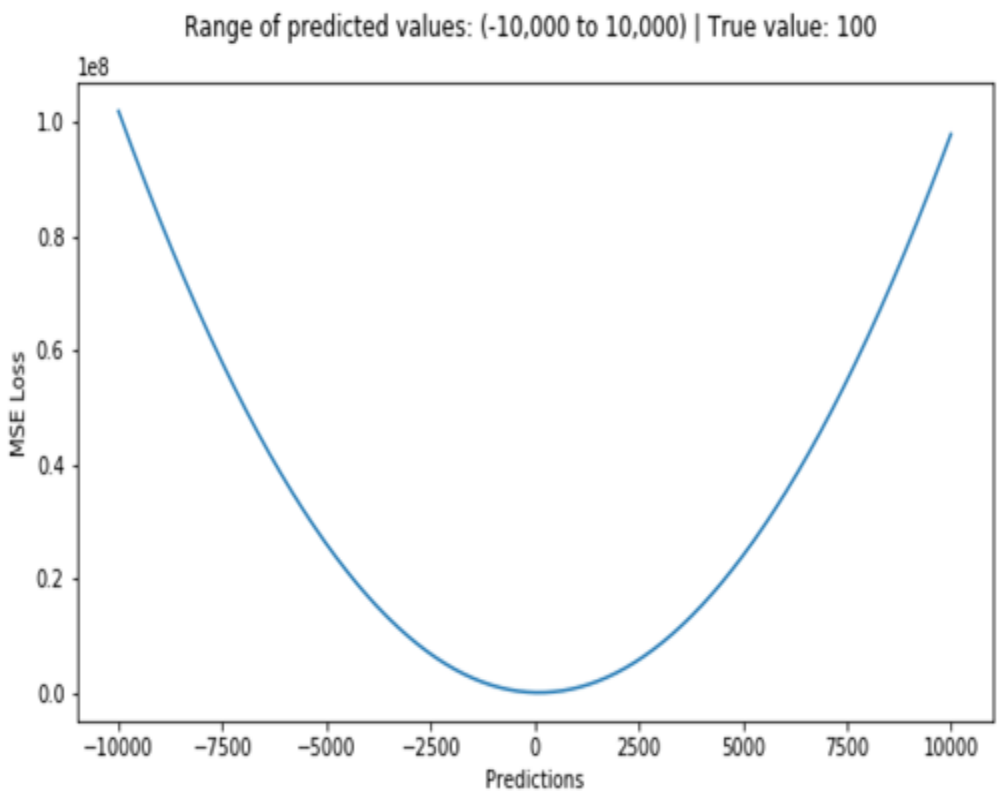
A simpler view point

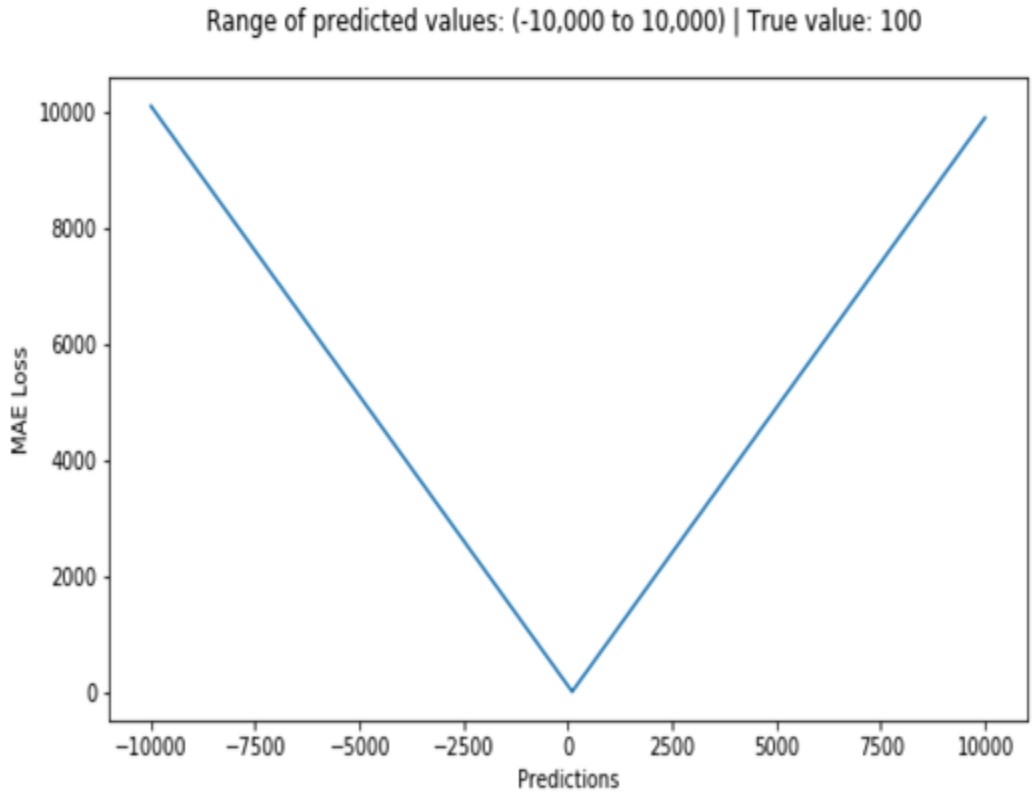


Blocks with Learnable parameters Matrix Multiplication

Nonlinear functions (often non learnable)

Loss Functions





Mean Squared Loss

$$L(y, y') = (y - y')^2$$

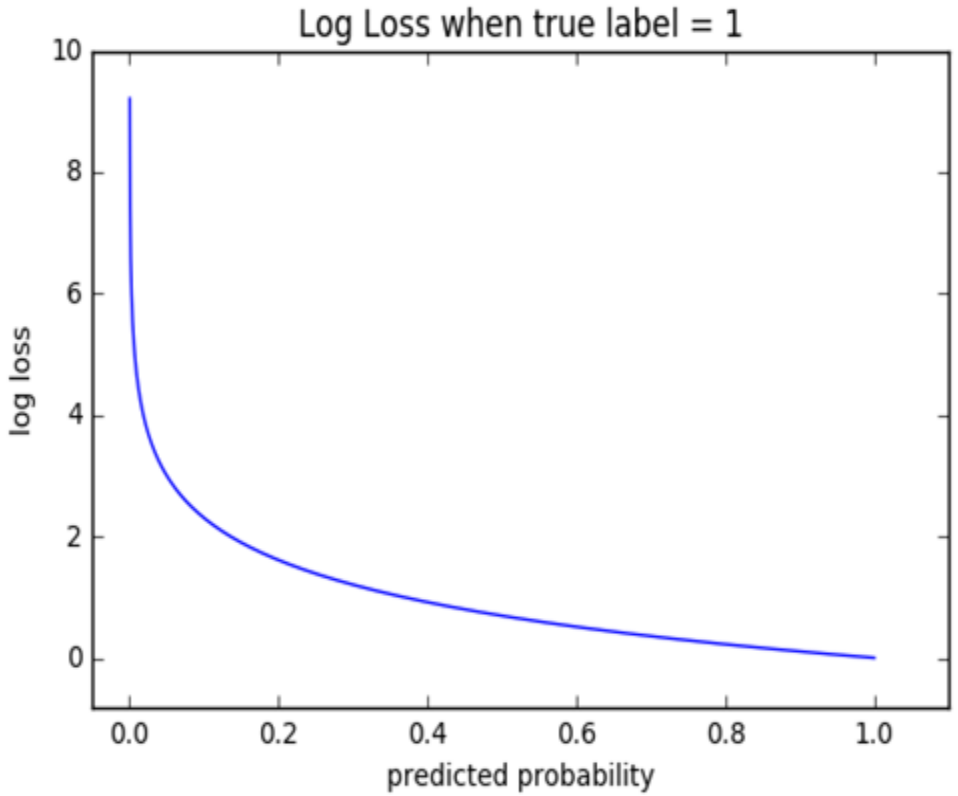
y - Actual value

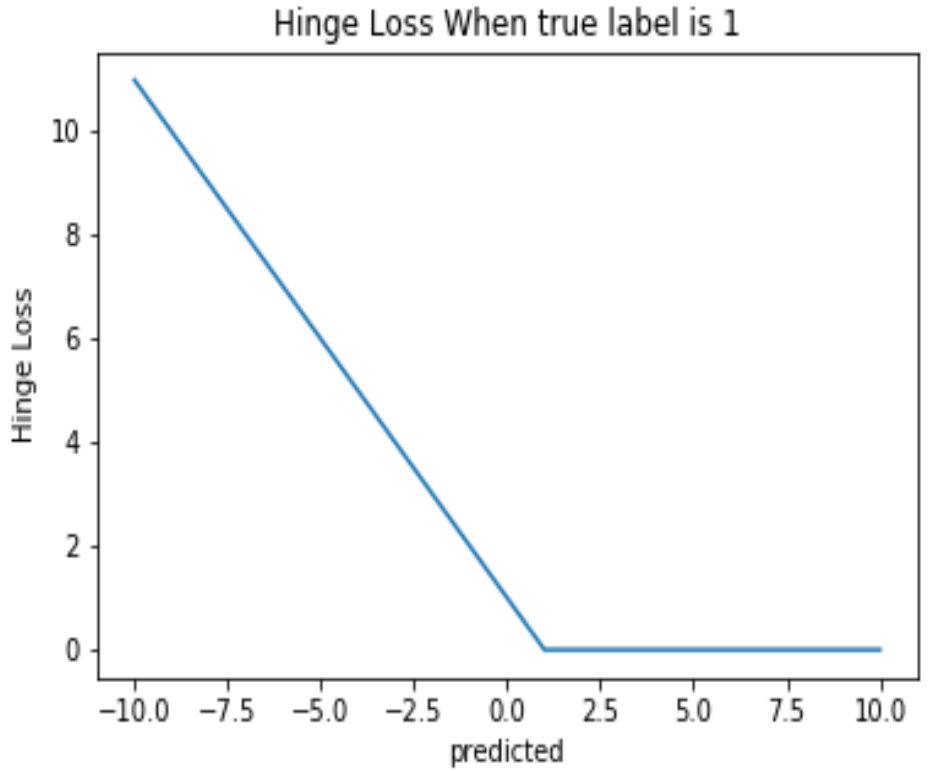
y' - Predicted value

Mean Absolute Loss

$$L(y, y') = |y - y'|$$

Loss Functions





Cross Entropy Loss

$$L(y, y') = -(y log(y') + (1 - y) log(1 - y'))$$

y, - Actual value

y' - Predicted value

Hinge Loss

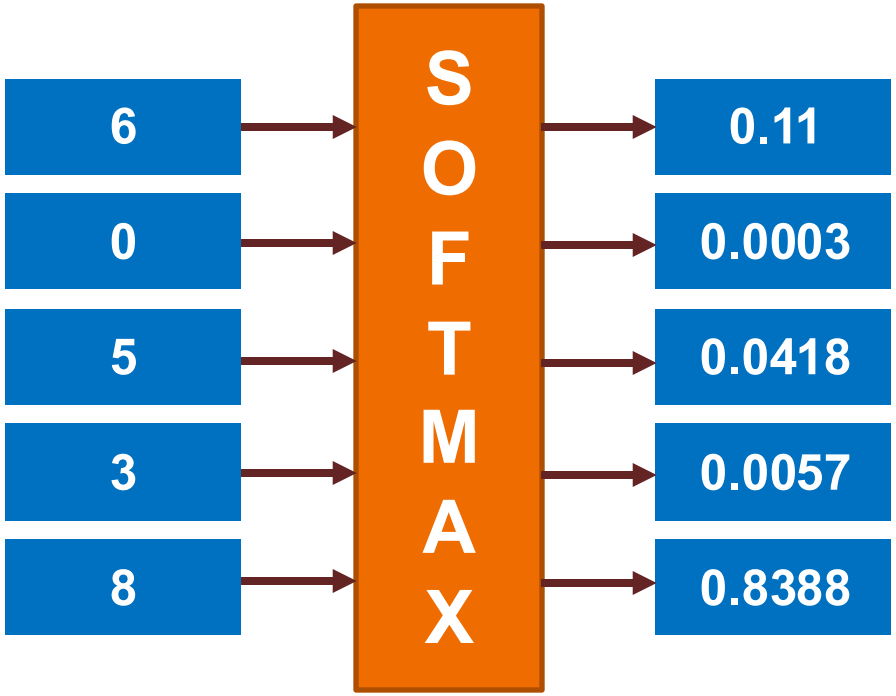
$$L(y, y') = max(0, 1 - y * y')$$

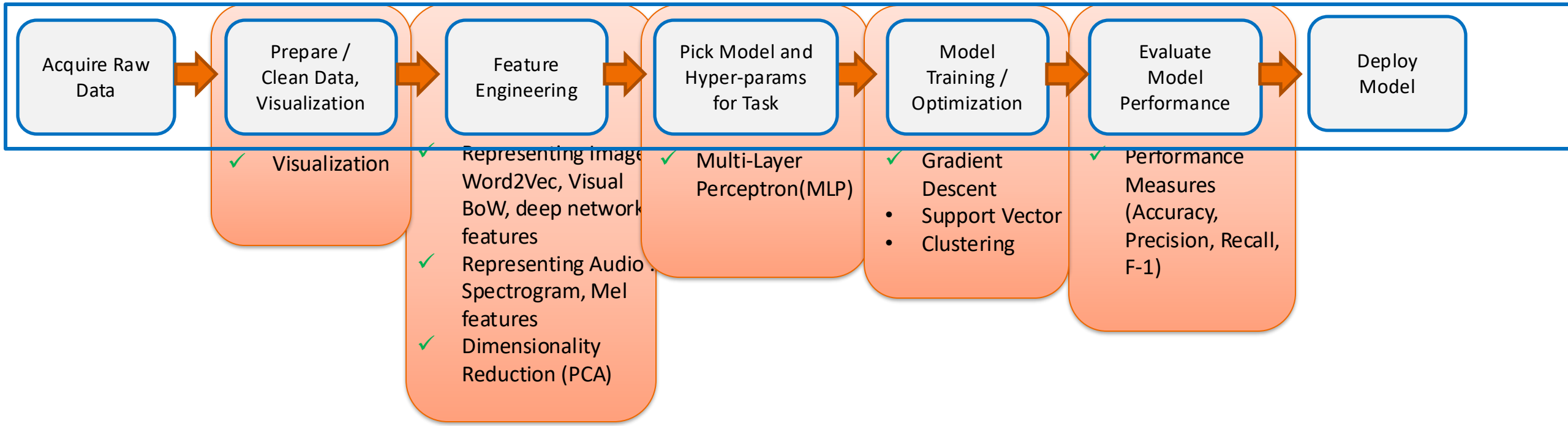
Softmax

```
Out[12]: array([ 6.,  0.,  5.,  3.,  8.])

In [8]:  exp = (np.e)**(x)
         exp
         executed in 6ms, finished 01:47:23 2018-08-21

Out[8]:  array([  4.03428793e+02,   1.00000000e+00,   1.48413159e+02,
                  2.00855369e+01,   2.98095799e+03])

In [9]:  sigma_e = np.sum(exp)
         sigma_e
         executed in 9ms, finished 01:47:25 2018-08-21

Out[9]:  3553.8854765602264

In [11]: z = exp/sigma_e
         z
         executed in 8ms, finished 01:47:34 2018-08-21

Out[11]: array([  1.13517669e-01,   2.81382168e-04,   4.17608165e-02,
                  5.65171192e-03,   8.38788421e-01])
```

- Normalizes the output.
- K is total number of classes

$$z_n = \frac{e^{x_n}}{\sum_{i=1}^{K} e^{x_i}}$$





SVMs and Kernels

Kernel as Similarity Function

Linear Classifier

- Decision boundary: Hyperplane

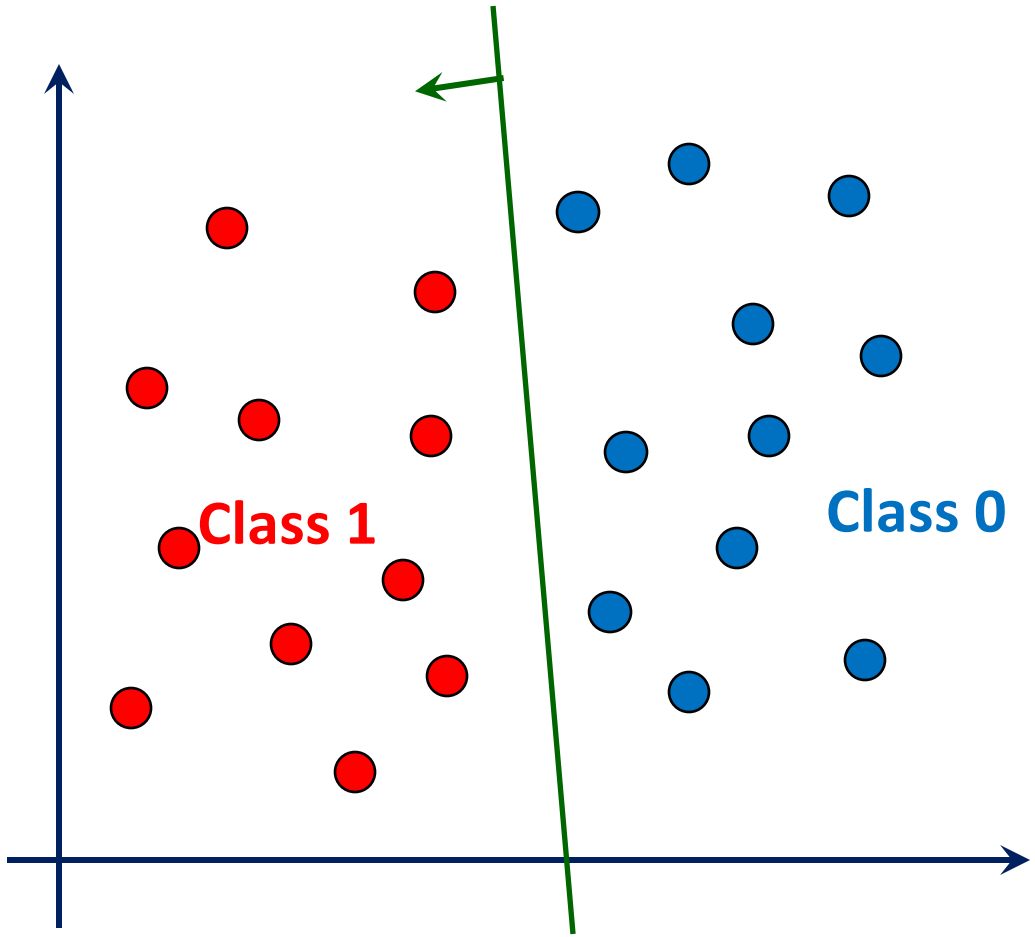
$$w^T x = 0$$

- Class 1 lies on the positive side

$$w^T x > 0$$

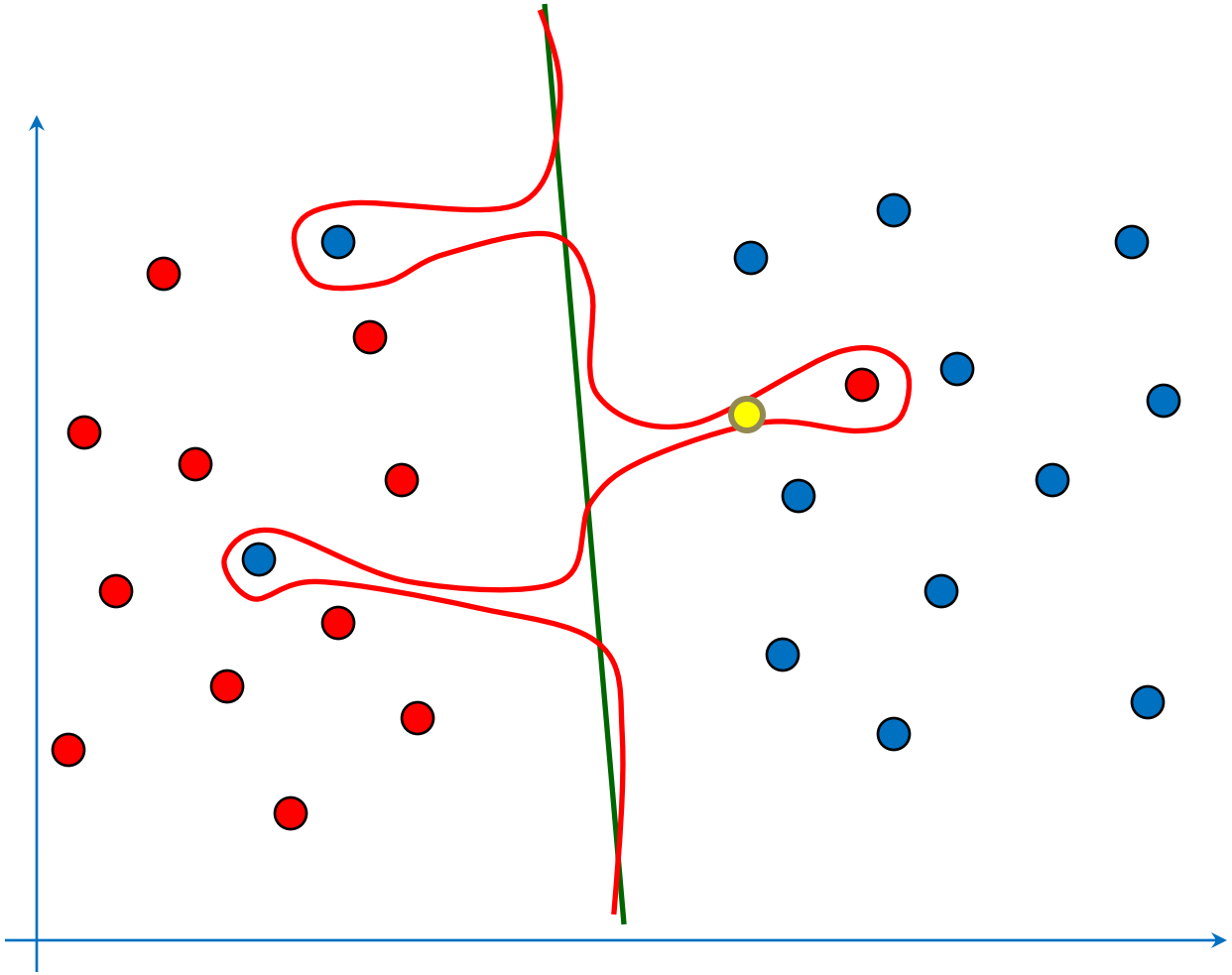
- Class 0 lies on the negative side

$$w^T x < 0$$



Why Linear? Generalization vs. Complexity

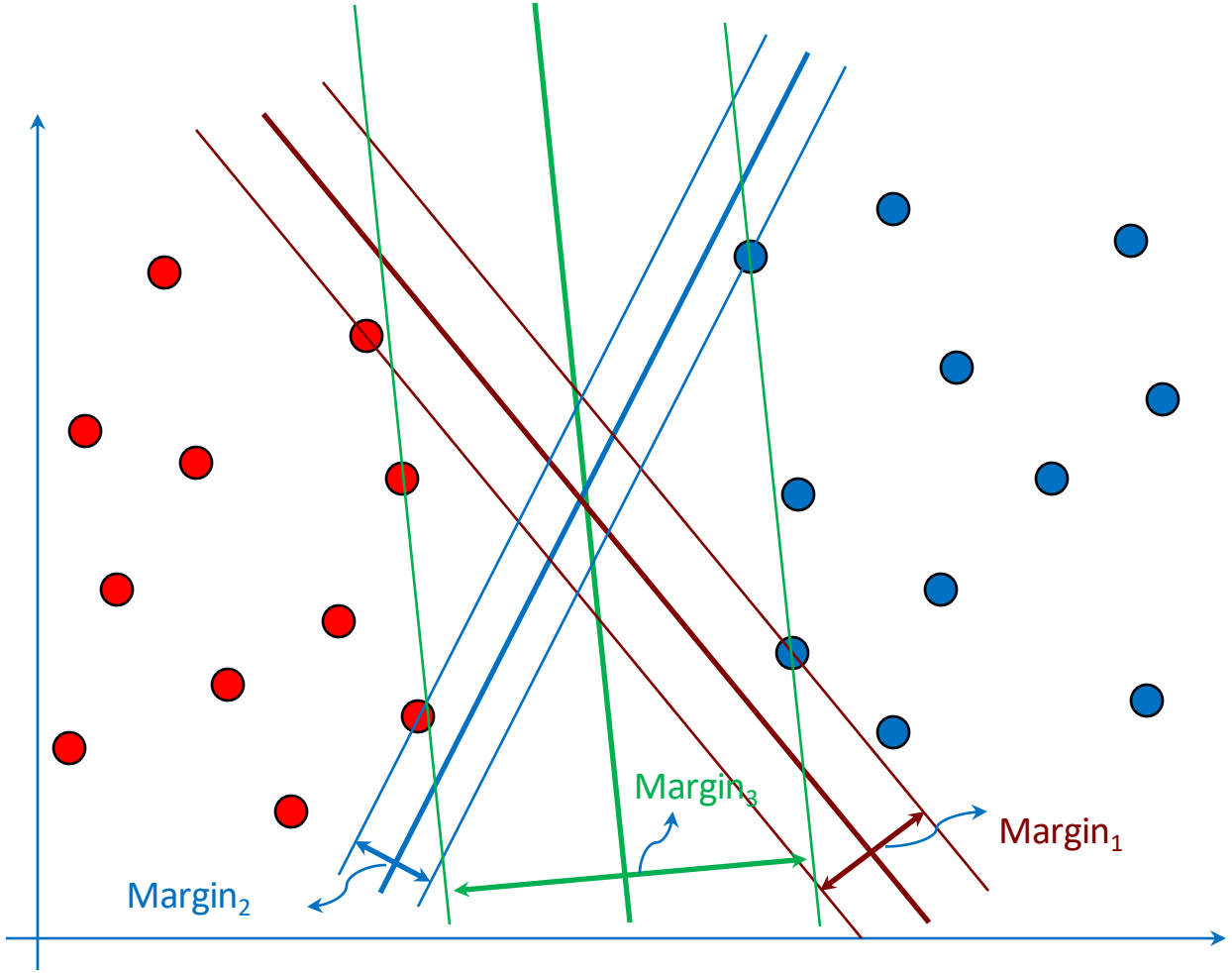
- Is it good to use a complex curve to reduce training error?



Are both solutions equally good?

Margin: The No-mans Band

- Margin: Width of a band around decision boundary without any training samples
- Margin varies with the position and orientation of the separating hyperplane

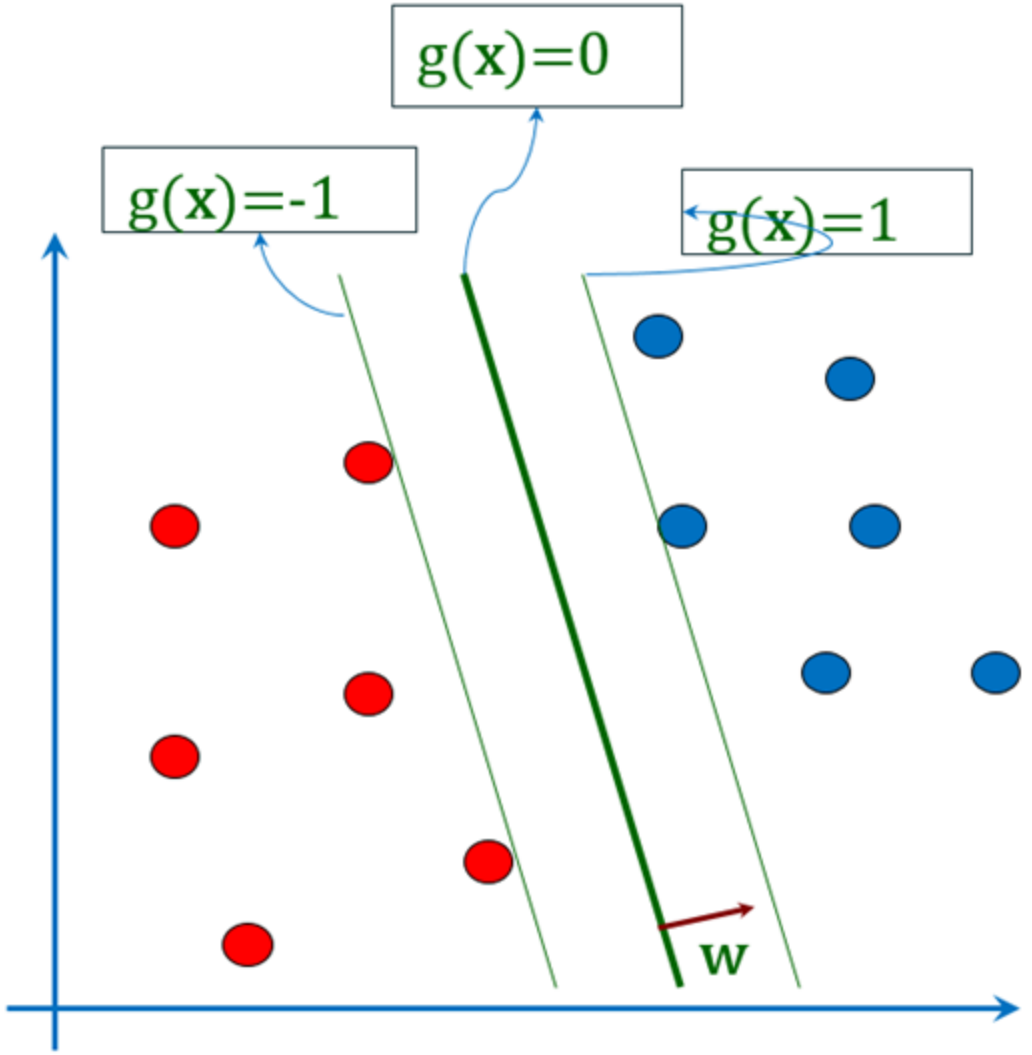


Is a Larger Margin better? Why?

SVM: Formulation

- Let $g(X) = W^T X + b$

- We want to maximize margin:
  - $W^T X_i + b \leq -1$ for $y_i = -1$
  - $W^T X_i + b \geq 1$ for $y_i = 1$
  - Or $y_i(W^T X_i + b) \geq 1$ for i.



SVM Optimization: Convex Optimization

$$min\ \frac{1}{2}W^TW$$

$$y_i(W^Tx_i + b) - 1 \geq 0 \forall i$$

$$y_i \epsilon \{1, -1\}$$

$$J_d(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2}\alpha_i \alpha j y_i y_j \boxed{x_i^T x_j} \sum_{i=1}^{N} \alpha_i y_i = 0$$

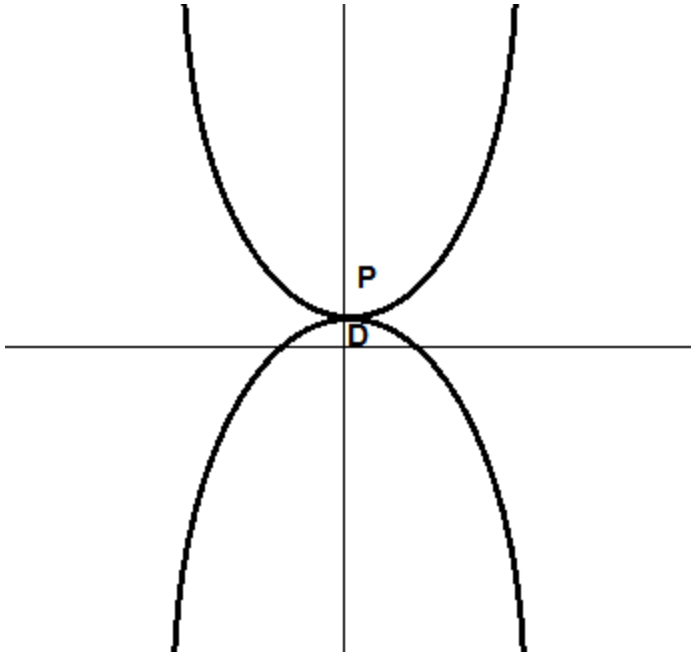
Convex Opt.
Only dot products

$$W = \sum_{i=1}^{N} \boxed{\alpha_i y_i x_i}$$

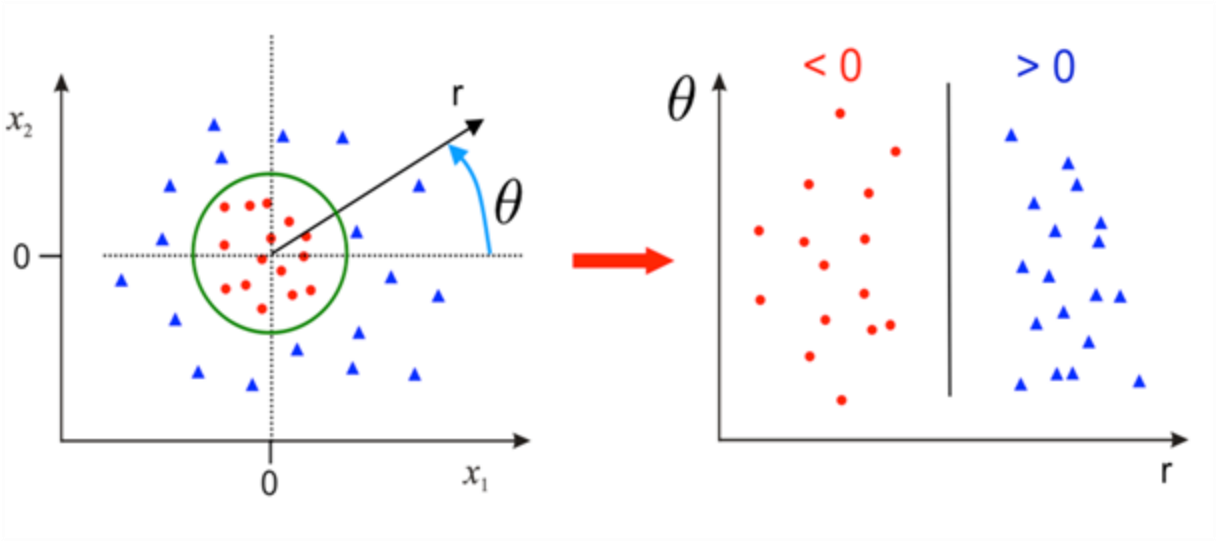
Support Vectors

$$W^TX = \sum^{N} \alpha_i y_i \boxed{x_i^T x}$$

Only dot products



Nonlinearity with Feature Maps

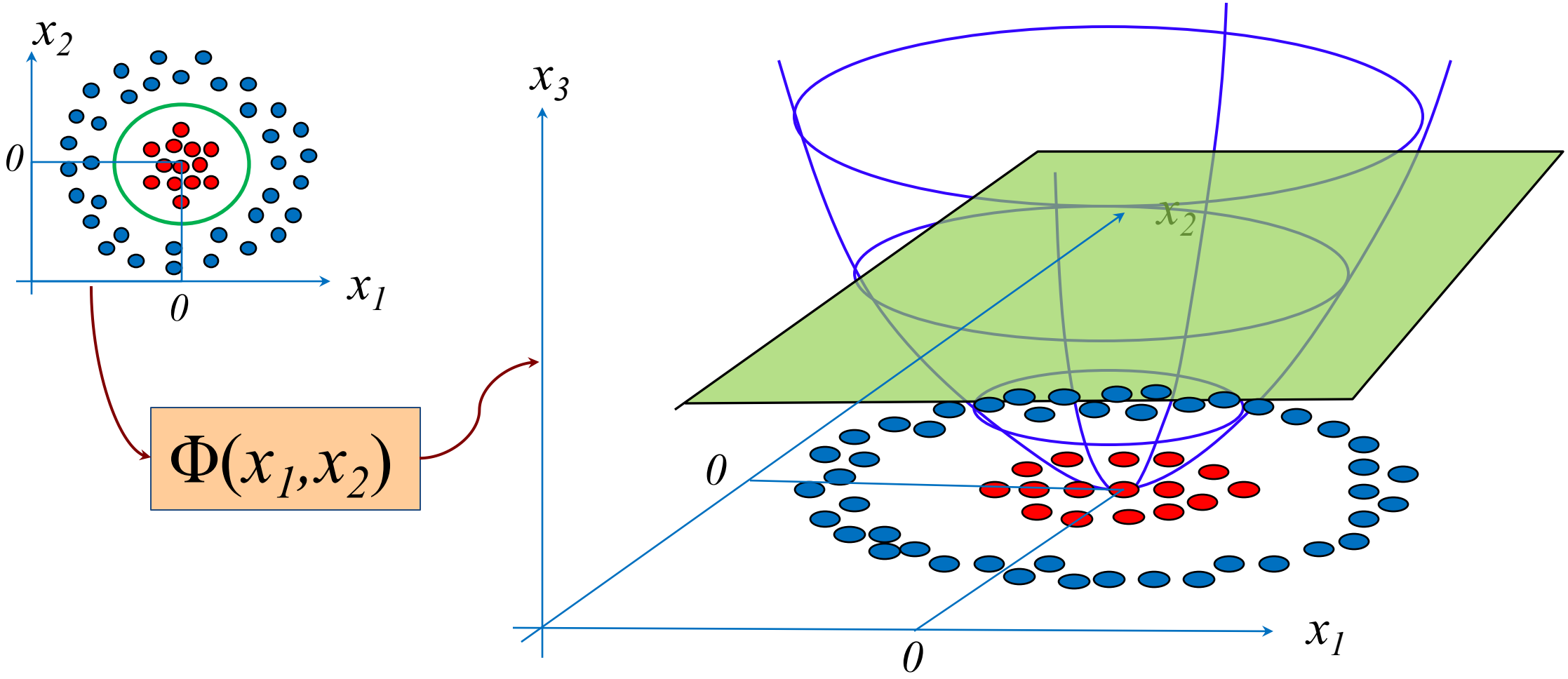


$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \rightarrow \begin{pmatrix} r \\ \theta \end{pmatrix}$$

- With a “smart” feature map, a linearly non-separable problem can be converted to a separable problem!!

Non-linear Mapping

$$x_3 = x_1^2 + x_2^2$$



Φ is a non-linear mapping into a possibly high-dimensional space

Kernel Strategy

$$w = \sum_{i=1}^{N} \alpha_i y_i x_i$$

What we need is only $w^T x = \sum_{i=1}^{N} \alpha_i y_i x_i^T x$

We can do the same in a new feature space:

$$w^T x = \sum_{i=1}^{N} \alpha_i y_i \phi(x_i)^T \phi(x) \qquad w^T x = \sum_{i=1}^{N} \alpha_i y_i K(x_i, x)$$

Clustering

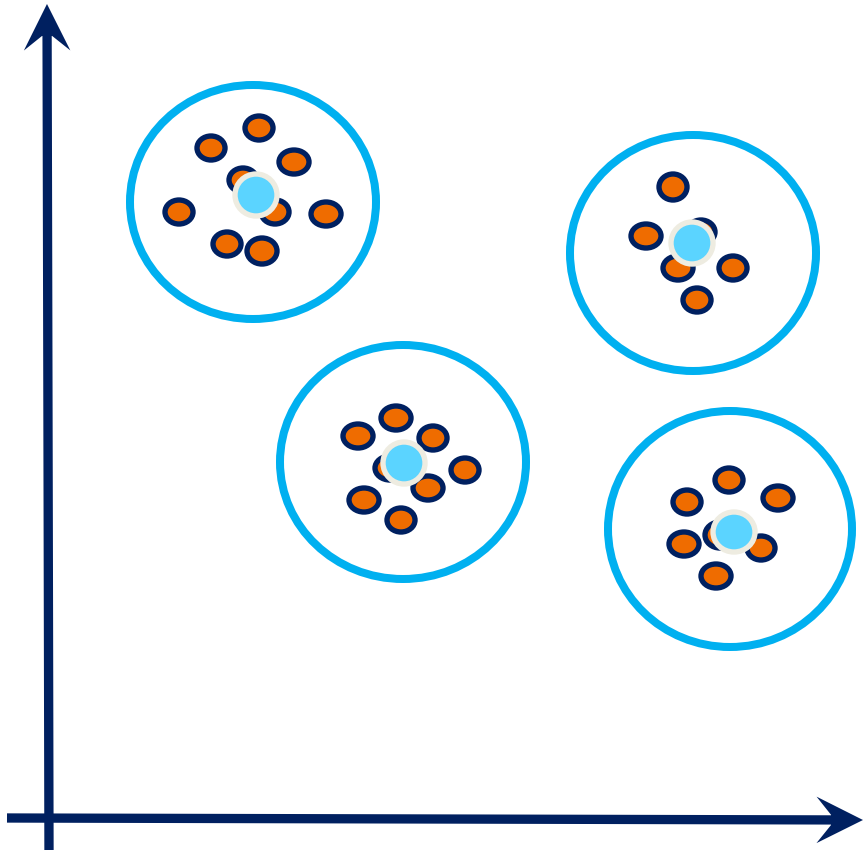
Identifying Similar Patterns

K-Means

- You are given N points

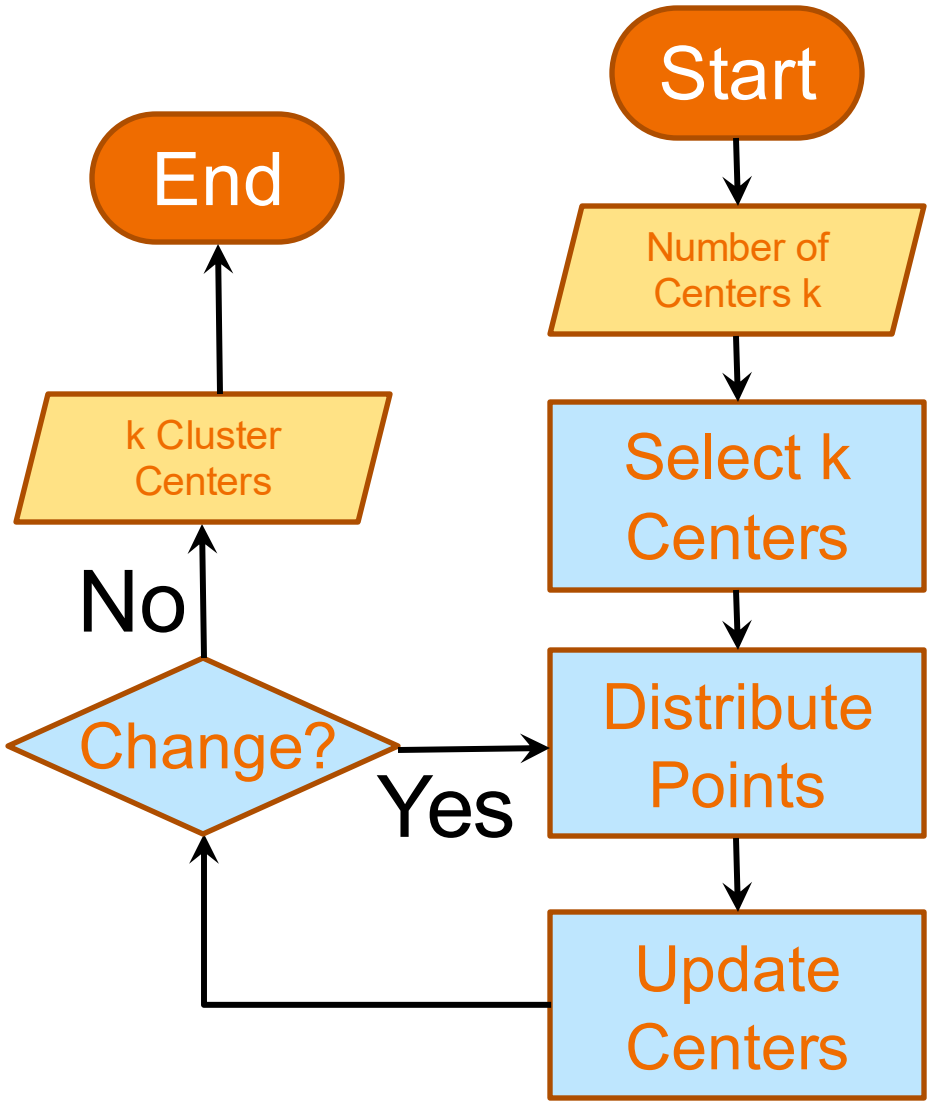
- How do we find k clusters?
  - What if we know the cluster centers?

- How do we find the cluster centers?
  - What if we know the k clusters?



K-Means

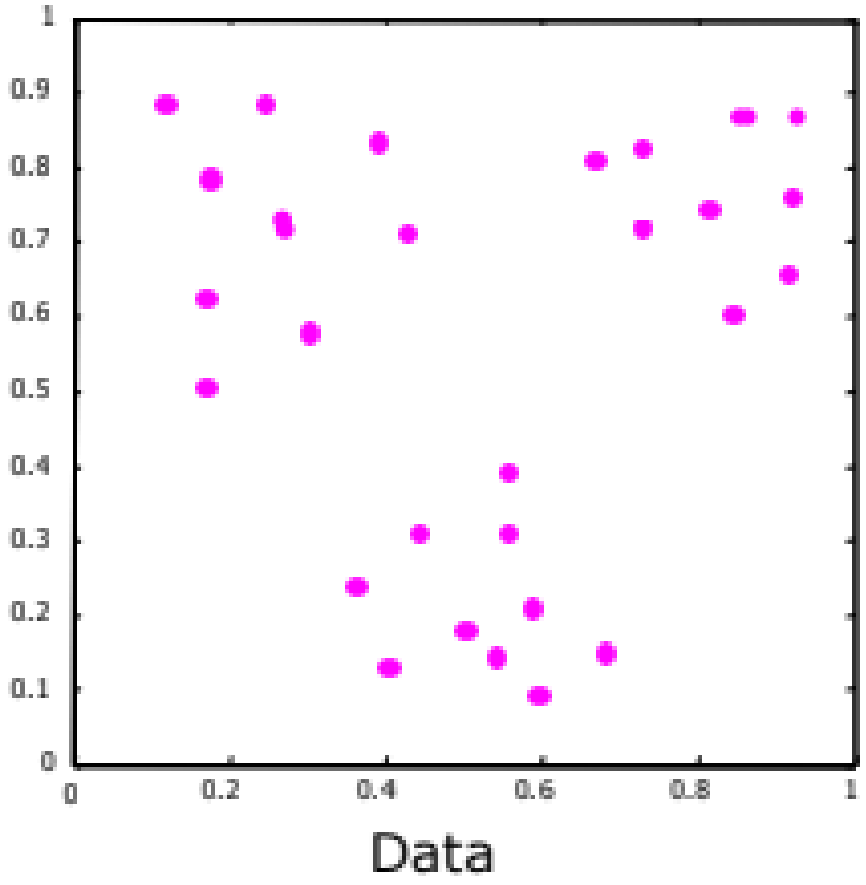
1. Input: k (number of clusters)
2. Randomly select k centers
3. Distribute Points
4. Update Centers
5. Repeat 3,4 till convergence
6. Output: Cluster centers

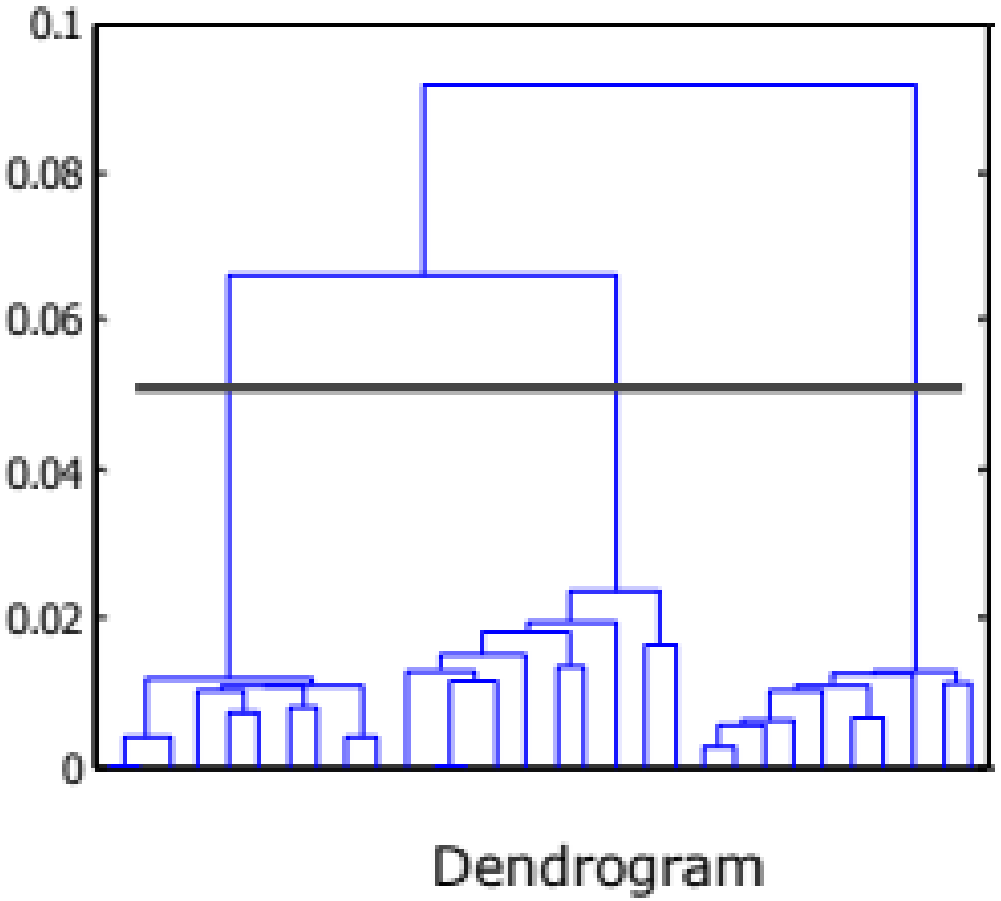


Single-Link Algorithm

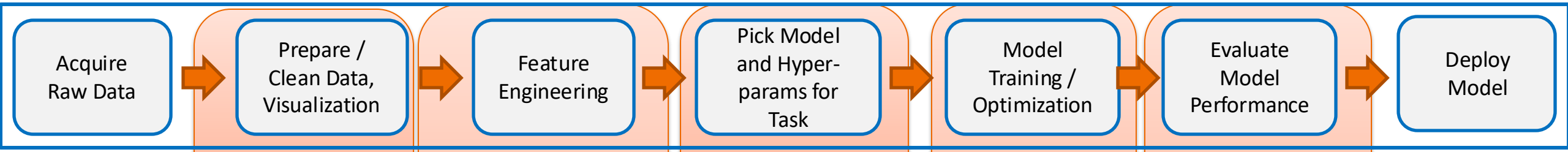
- Form a hierarchy for the data points (dendrogram), which can be used to partition the data
- The “closest” data points are joined to form a cluster at each step

Single-Link Algorithm





Summary



- ✓ Visualization
- ✓ Representing Image: Word2Vec, Visual BoW, deep network features
- ✓ Representing Audio : Spectrogram, Mel features
- ✓ Dimensionality Reduction (PCA)
- ✓ Multi-Layer Perceptron(MLP)
- ✓ Gradient Descent
- ✓ Support Vector machines
- ✓ Clustering
- ✓ Performance Measures (Accuracy, Precision, Recall, F-1)

Thanks!!

Questions?